Bachelor of Science in Computer Science & Engineering



Threat-Aware Waste Detection and Classification in Real-World Condition Using Computer Vision

by

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July, 2026

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Submitted in partial fulfilment of the requirements for
Degree of Bachelor of Science
in Computer Science & Engineering

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The thesis titled '**Threat-Aware Waste Detection and Classification in Real-World Condition Using Computer Vision**' submitted by ID: 2004083, Session 2023-2024 has been accepted as satisfactory in fulfilment of the requirement for the degree of Bachelor of Science in Computer Science & Engineering to be awarded by the Chittagong University of Engineering & Technology (CUET).

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Acknowledgements

First and foremost, I express my deepest gratitude to the Almighty for granting me the strength, patience, and ability to complete this work.

I am profoundly grateful to my supervisor, Associate Professor Mohammad Obaidur Rahman, Department of Computer Science & Engineering, CUET, for his invaluable guidance, encouragement, and patience throughout this research. His insightful feedback and willingness to discuss ideas at every stage shaped this thesis and greatly improved my understanding of the subject.

I would also like to thank the faculty members and staff of the Department of Computer Science & Engineering, CUET, for their support and for providing the academic environment in which this work was carried out.

Finally, I am deeply thankful to my family for their constant support, sacrifices, and encouragement, without which this journey would not have been possible.

Abstract

Effective waste management depends on correctly identifying an item's material, condition, disposal route, and handling threat. Existing computer-vision approaches largely emphasize material classification, leaving safe disposal under real-world conditions less studied. To address this gap, this thesis proposes a threat-aware, two-stage waste detection and classification system. A single-class YOLO11s detector localizes the item; a shared ConvNeXt-S backbone then predicts material (ten classes), condition (Good/Poor), route (Reuse, Recycle, Landfill, Special), and ordinal threat (0–3) using CORN. Calibration and regulation-grounded routing are evaluated alongside classification, with rules that allow predicted hazards to trigger special handling. Ten classifier configurations spanning convolutional networks, transformers, and an ensemble were compared, with ConvNeXt-S strongest among the individual backbones on material classification. Cross-validated evaluation over 457 items yields material accuracy 0.832 (macro-F1 0.803), threat QWK 0.771, condition AUROC 0.821, route macro-F1 0.675, and material ECE 0.036. The detector reaches mAP@50 0.992. On 50 test items, oracle and detector crops both yield material accuracy 0.840 at a detection rate of 1.0, with small decreases in condition and route performance. The routing policy sends 148 of 169 hazardous items to Special (recall 0.876); including two referrals for review raises safety recall to 0.888 at coverage 0.961. External validation and rare-class analysis identify remaining visual ambiguities, routing errors, and transfer limitations.

Keywords: Waste Classification, Object Detection, Multi-Task Learning, Ordinal Regression, Threat-Awareness, Calibration, YOLO, ConvNeXt, Computer Vision.

Table of Contents

Acknowledgements **iii**

Abstract **iv**

List of Figures **viii**

List of Tables **x**

1 Introduction **1**

1.1 Introduction 1
1.2 Difficulties 3
1.3 Applications 3
1.4 Motivation 4
1.5 Contribution of the thesis 5
1.6 Thesis Organization 5
1.7 Conclusion 6

2 Literature Review **7**

2.1 Introduction 7
2.2 Deep Learning for Waste Material Classification 7
2.3 Object Detection for Waste in Context 8
2.4 Public Datasets for Waste Recognition 9
2.5 Two-Stage and Multi-Signal Systems 10
2.6 Hazardous and Safety-Aware Waste Handling 11
2.7 Ordinal Regression and Model Calibration 11
2.8 Summary of Related Work 12
2.9 Limitations of Existing Works 12
2.10 Terminologies 13
2.11 Conclusion 13

3 Methodology **15**

3.1 Introduction 15
3.2 System Overview 15
3.3 Experimental Setup 18
3.4 Methodology Workflow 18

3.5 Dataset Development 20
3.5.1 Acquisition 20
3.5.2 Labelling Scheme 21
3.5.3 Preprocessing 24
3.5.4 Augmentation and Class Balancing 25
3.5.5 Item-Level Split and Manifests 26
3.5.6 Class Distribution 26
3.6 Detection Stage 27
3.6.1 Detector Families Considered 27
3.6.2 YOLO11 Architecture 27
3.6.3 Why YOLO11 28
3.6.4 Detector Training and Inference 30
3.7 Proposed Multi-Task Network 30
3.7.1 Backbone Candidates 30
3.7.2 ConvNeXt-S Architecture 30
3.7.3 Why ConvNeXt-S 32
3.7.4 Vision-Transformer Fine-Tuning and Feature Fusion . . . 32
3.7.5 Task Heads 33
3.8 Loss Functions 33
3.9 Training Procedure 35
3.10 Calibration 36
3.11 Decision Layer 37
3.12 Evaluation Protocol 38
3.13 Conclusion 39

4 Results and Discussions 40
4.1 Introduction 40
4.2 Performance Evaluation Metrics 40
4.2.1 Precision, Recall, and F1-Score 40
4.2.2 mAP@50 and mAP@50-95 (detection) 41
4.2.3 Quadratic Weighted Kappa (threat) 41
4.2.4 AUROC, ECE, and Hazard Recall 41
4.3 Classification Backbone Benchmark 41
4.4 Headline Results: Deployed Model and Cross-Validated Robustness 44
4.4.1 Per-Class Material Performance 46
4.4.2 Confusion Matrices and ROC Curves 47
4.4.3 Calibration and Training Behaviour 48
4.5 Detection and End-to-End Performance 49
4.5.1 Detector Benchmark 49

4.5.2 Deployed Detector and End-to-End Cost 50
4.6 External Validation 51
4.7 Multi-View Aggregation 53
4.8 Interpretability and Qualitative Results 54
4.9 Rare-Class Analysis 55
4.10 Discussion and Comparison with Related Work 57
4.11 Conclusion 58

5 Conclusion **60**
5.1 Conclusion 60
5.2 Limitations and Threats to Validity 61
5.3 Future Work 61

List of Figures

1.1 High-level overview of the proposed threat-aware, two-stage waste detection and classification system. 2

3.1 End-to-end inference pipeline of the proposed system. 16
3.2 Working-process flowchart of the complete system, from the input photograph to the final routing decision, exposing the decision-layer branches and the threat-aware safety backstop. 17
3.3 Training, validation, and testing workflow of the two-stage system. 19
3.4 Dataset development flow from acquisition to leakage-checked splits. 20
3.5 Labelling scheme: the four annotation dimensions of every item. . 21
3.6 The labelling scheme applied to one real item (item 106, broken glass). 22
3.7 Classifier preprocessing chain. 24
3.8 Both preprocessing paths applied to one real item (item 106). . . 25
3.9 Class distribution of the dataset across materials, disposal routes, and threat levels. 26
3.10 Internal architecture of the YOLO11 detector. 29
3.11 Internal architecture of the ConvNeXt-S backbone. 31
3.12 The full detect → classify inference on one real item (item 106). . 34
3.13 Stages of the decision layer. 38

4.1 Backbone comparison per head. 44
4.2 Deployed ConvNeXt-S versus the four-backbone fusion: learned-state size and per-head performance differences. Fusion gains 0.031 threat QWK with 3.76× the learned-state elements; inference latency remains unmeasured. 44
4.3 Cross-validated metrics (457-item out-of-fold pool, 95% confidence intervals) with the deployed model's 50-item test scores overlaid. . 45
4.4 Cross-validated confusion matrices for the four task heads. 47
4.5 ROC curves for the condition and route heads. 48
4.6 Cross-validated material reliability and per-head ECE before/after temperature scaling of the earlier three-backbone fusion. 49
4.7 Multi-task training loss and per-head validation metrics of ConvNeXt-S. 49

4.8 Earlier YOLO11 detector evaluation: precision–recall, F1–confidence, precision–confidence, and recall–confidence curves. . . 52
4.9 End-to-end material accuracy versus an oracle crop. 52
4.10 Material transfer to TrashNet under restricted four-way zero-shot and six-way fine-tuned evaluation. 53
4.11 Multi-view aggregation: material macro-F1 and threat QWK (left), and threat recall versus false-positive rate at level ≥ 2 (right). Numbered labels identify the aggregation methods. 54
4.12 Grad-CAM examples: original image, material-head saliency with view confidence, and threat evidence. 56
4.13 End-to-end pipeline walkthrough: detection, crop, and the final routing decision. 57

List of Tables

2.1 Representative public datasets for waste recognition. 10
2.2 Summary of representative related work. 12

3.1 Label spaces of the four prediction tasks. 22
3.2 Threat severity (Litter Impact Index) [25]. 23
3.3 Routing matrix: material × condition → route. 23
3.4 Detector candidates and their defining architectural idea. 28
3.5 Backbone candidates evaluated as the shared trunk. 31
3.6 Training hyperparameters. 36

4.1 Per-backbone results on the held-out split (item-level); best per column in bold. 42
4.2 Headline metrics of ConvNeXt-S and the cross-validated robustness check. Classifier test metrics use oracle crops; decision test metrics use the integrated pipeline. 45
4.3 Cross-validated per-class material F1 and support (457-item evaluation). 46
4.4 Detector comparison on the held-out test split; best per column in bold. 50
4.5 Separate-model outcomes versus the integrated end-to-end pipeline. 51
4.6 Multi-view aggregation on 457 cross-validated items, with mode threat decoding. Threat recall and FPR use threshold ≥ 2; best per column in bold. 54
4.7 Cross-validated confusion for the two weakest material classes. . . 57
4.8 Comparison with published reference points. 58

Chapter 1

Introduction

1.1 Introduction

The management of solid waste is one of the most pressing environmental and public-health challenges of the modern era. According to the World Bank, the world generated roughly 2.01 billion tonnes of municipal solid waste in 2016, a figure projected to rise to about 3.40 billion tonnes by 2050 as populations grow and consumption intensifies [1]. A large and increasing share of this stream is mixed, contaminated, and hazardous, while the sorting infrastructure that must process it, particularly in rapidly urbanizing countries such as Bangladesh, still relies heavily on manual labour. Correct sorting at the point of disposal determines whether a material is recovered for reuse or recycling, sent to landfill, or routed for special hazardous handling; errors are costly both economically, through contaminated and therefore worthless recycling streams, and in terms of safety, when a battery, a medical sharp, or a chemical container is handled as ordinary trash.

Manual sorting is slow, expensive, and, most seriously, dangerous. Workers are routinely exposed to sharp objects, leaking chemicals, and lithium-ion batteries that can ignite, and the cognitive load of distinguishing dozens of material and hazard categories by eye leads to inevitable errors. These pressures motivate automation, and computer vision is the natural enabling technology: a camera observing a disposal point or conveyor can, in principle, recognize what an item is, judge its physical condition, and decide where it should go, far faster and more consistently than a human, and without placing a person in harm's way.

However, most existing vision systems for waste address only a fragment of this

problem. They reduce the task to material classification on clean, well-lit, studio-style photographs, and they stop at a coarse "recyclable versus non-recyclable" label. Two dimensions that matter enormously for a deployable system are typically missing. The first is the real-world condition of the item: deployment images contain soiling, breakage, deformation, occlusion, background clutter, and uneven lighting that curated benchmarks do not, and an item's condition (for example, intact versus broken glass) can itself change the correct disposal route. The second, and more critical, is the threat an item poses to the people and machinery that must handle it, information that is essential for safe routing but is absent from material-only systems.

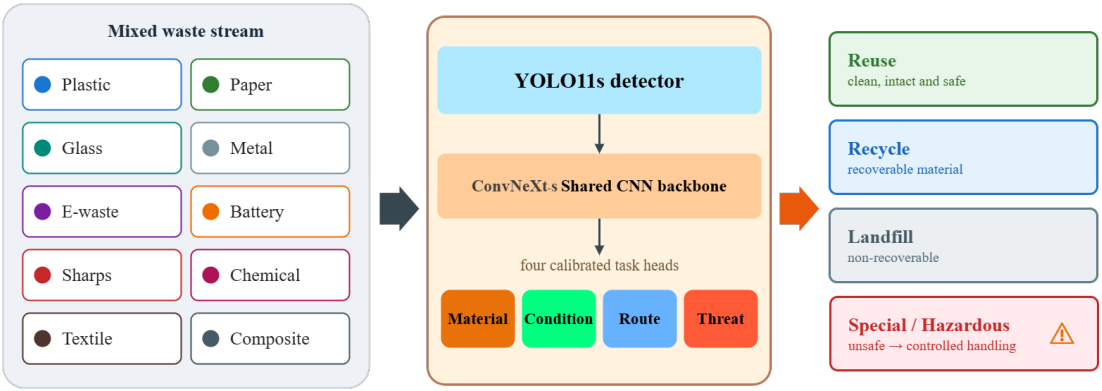


Figure 1.1: High-level overview of the proposed threat-aware, two-stage waste detection and classification system.

This thesis addresses both gaps directly; Figure 1.1 gives a high-level overview of the resulting system. We propose a threat-aware, two-stage detection-and-classification system that operates on real-world-condition images. The system first detects and crops the waste item, then, from a single shared representation, jointly predicts its material across ten classes, its condition, its disposal route, and an ordinal threat level on a four-point severity scale. These four predictions are reconciled by a decision layer grounded in published waste-management regulation and equipped with an explicit safety backstop, so that an item judged to be highly hazardous is routed for special handling even when its material label is visually ambiguous. The result is a complete pipeline (detection, multi-task classification, calibration analysis, and a safety-aware decision) rather than an isolated classifier, and it is evaluated through classification, localization, and routing performance under challenging capture conditions.

1.2 Difficulties

Several challenges were encountered during this research:

- **Real-world image variability:** Unlike curated datasets, the images used here contain genuine variation in lighting, background clutter, occlusion, viewing angle, and physical wear. A robust system must generalize across all of these rather than memorizing clean exemplars.
- **Visual ambiguity of safety-critical classes:** Some of the most important classes are intrinsically hard to separate from appearance alone. A chemical container can look identical to an ordinary plastic or glass bottle, and a composite (multi-material) item visually resembles whichever component is most prominent. These ambiguities make appearance-only classification particularly difficult for the rarest classes.
- **Severe class imbalance and limited data:** The taxonomy includes both common streams (e.g., plastic) and scarce but critical ones (e.g., hazardous chemical, composite), with only a few dozen items in the smallest classes. Training and, especially, evaluating such rare classes reliably is difficult.
- **Modelling threat as an ordered quantity:** Threat severity is not a set of unrelated categories but an ordered scale; a model should be penalized more for confusing "low" with "high" than "low" with "moderate." This requires an ordinal formulation rather than ordinary classification.
- **Computational constraints:** Training nine deep backbones, six object detectors, and a cross-validation study under a fixed GPU budget required careful, efficient experimental design.

1.3 Applications

- **Automated and assisted sorting:** The system is intended to support a sorting station (on a conveyor, at a collection point, or as a phone/edge application) by recognizing the item and recommending a disposal route.

- **Worker and public safety:** By explicitly predicting threat level and directing predicted hazardous items to special handling, the system targets the most dangerous and most under-studied part of waste sorting: sharps, batteries, electronic waste, and chemicals.
- **Recycling-stream quality:** Correct material and condition prediction can reduce contamination of recyclable streams, improving the value and recoverability of collected material.
- **Decision support and auditing:** Because the decision layer is rule-based and produces an explanation and a confidence value for each item, its outputs can be audited and deferred to a human when uncertain.
- **Regulatory alignment:** Routing is grounded in published waste-management regulations, so the system's recommendations map to actual local disposal streams rather than abstract labels.

1.4 Motivation

The motivation for this research arises from the following considerations:

- **Safety first:** Mis-handling hazardous waste endangers sanitation workers and the public. A system that treats threat as a first-class output and provides a prediction-based safety backstop directly addresses a problem that material-only classifiers ignore.
- **Realism over benchmarks:** Studio-condition accuracy does not necessarily transfer to deployment. Training and evaluating under genuine real-world condition is necessary for the results to be meaningful.
- **Trustworthy automation:** For automation to be adopted, its confidences must be reliable and it must be able to abstain. Calibration analysis and a confidence-aware decision layer address these requirements.
- **Local relevance:** Grounding the routing rules in Bangladesh's solid-waste regulations and the national 3R strategy makes the system relevant to the context in which it would be deployed.

1.5 Contribution of the thesis

This thesis makes the following contributions:

- It formulates waste handling as a four-head multi-task problem (material, condition, route, and an ordinal threat level) on a shared backbone, rather than as isolated material classification. To our knowledge, modelling disposal-threat severity as a rank-consistent ordinal head wired into a decision cost is novel in the waste computer-vision literature.
- It implements a complete two-stage detect-then-classify pipeline (YOLO11s detector followed by the multi-task classifier) and shows no measurable drop in material accuracy versus an oracle crop on 50 test items (accuracy 0.840 for both, at a detection rate of 1.0), although condition and route scores decrease slightly.
- It introduces a regulation-grounded decision layer with confidence-aware reconciliation and abstention, and a threat-based safety backstop that directs predicted hazardous items to special handling regardless of their predicted material. Per-head temperature calibration is evaluated to assess confidence quality.
- It provides a rigorous evaluation: a comparison of ten classifier configurations spanning convolutional and transformer backbones and an ensemble, cross-validated evaluation over 457 items with bootstrap 95% confidence intervals, calibration analysis, paired significance testing, external validation on public datasets, and an honest, evidence-based analysis of the hardest rare classes.

1.6 Thesis Organization

The remainder of this report is organized as follows:

- **Chapter 2** reviews related work on waste detection and classification, threat/hazard handling, ordinal regression, and calibration, and identifies the gaps this thesis addresses.

- **Chapter 3** describes the proposed methodology, from dataset construction and the labelling scheme through the detector, the multi-task network, the loss functions, the training recipe, and the decision layer.
- **Chapter 4** presents the experimental results and analysis, including the backbone ablation, the cross-validated headline results, detection and external validation, and the rare-class analysis.
- **Chapter 5** concludes the thesis, summarizing the findings and outlining limitations and directions for future work.

1.7 Conclusion

This chapter introduced the problem of threat-aware waste detection and classification under real-world condition, motivated the need to model both item condition and handling threat alongside material and route, and summarized the difficulties, applications, and contributions of the work. The following chapter situates the proposed system within the existing literature.

Chapter 2

Literature Review

2.1 Introduction

Automated waste recognition has progressed from hand-engineered image-processing pipelines to end-to-end deep learning, and from coarse two-way recyclability decisions to fine-grained material taxonomies and in-context detection. This chapter surveys that progression along the dimensions most relevant to the present work. Section 2 reviews deep learning for waste material classification and the benchmark datasets that anchor it. Section 3 examines object detection for waste in cluttered scenes and the evolution of the single-stage detectors on which this thesis builds. Section 4 catalogues the public datasets that define the field. Section 5 discusses two-stage and multi-signal systems, the architectural family this work belongs to. Section 6 turns to the comparatively neglected problem of hazardous and safety-aware handling, and Section 7 to the modelling tools, ordinal regression and calibration, that this thesis uses to treat threat as a first-class output. Section 8 summarizes the reviewed works, Section 9 distills their common limitations into the research gap, and Section 10 defines the terminology used throughout the report.

2.2 Deep Learning for Waste Material Classification

The classification of a waste item by its constituent material is the most studied sub-problem in the field. Early systems relied on hand-crafted descriptors (colour histograms in RGB or HSV space, texture descriptors such as Local Binary Patterns, and shape features) fed to classical classifiers like Support Vector Machines.

While interpretable and cheap, such pipelines are brittle under the lighting, soiling, and background variation of real disposal scenes, and they require careful manual feature engineering for each new material.

The shift to convolutional neural networks (CNNs) removed the need for handcrafted features by learning hierarchical representations directly from pixels. Deep residual networks (ResNet) [2] made very deep models trainable through skip connections and remain a strong baseline; EfficientNet [3] achieved comparable accuracy at a fraction of the parameters through principled compound scaling; and MobileNetV3 [4] targeted mobile and edge deployment, which is directly relevant to a fielded sorting aid. More recently, ConvNeXt [5] modernized the pure-convolutional design to match transformer accuracy, and Vision Transformers (ViT) [6], together with self-supervised variants such as DINOv2 [7], have shown that attention-based and label-free pretraining can yield highly transferable features. This thesis evaluates representatives of all of these families as interchangeable backbones.

In the specific domain of waste, transfer learning from ImageNet-pretrained backbones is the dominant recipe. On the six-class TrashNet benchmark [8], fine-tuned networks reach very high accuracy: reported figures include GoogLeNet around 97.9% and ResNet50-based pipelines around 87%. On the larger and more varied twelve-class Garbage Classification dataset [9], EfficientNet-B0 reaches roughly 86% and ResNet50 roughly 88%. These numbers bracket what a material classifier should achieve, but they are obtained on clean, well-lit, single-object images; accuracy on genuinely cluttered or degraded photographs is typically lower, and is rarely reported. Hybrid approaches that combine deep features with classical machine-learning classifiers have also been explored to squeeze additional accuracy from limited data [10].

2.3 Object Detection for Waste in Context

Classification assumes the object has already been isolated; in practice a system must first locate the item. Object detectors fall into two families. Two-stage detectors (R-CNN and its successors) first propose candidate regions and then

classify them, achieving high accuracy at the cost of speed. Single-stage detectors predict boxes and classes in one pass and are therefore far faster, which is essential for real-time sorting.

The YOLO family epitomizes the single-stage approach. Since the original formulation that cast detection as a single regression problem [11], successive versions have steadily improved the accuracy–speed trade-off: YOLOv3 [12] introduced a stronger backbone and multi-scale prediction, and subsequent Ultralytics releases culminating in YOLO11 [13] added anchor-free detection, improved feature aggregation, and streamlined training. Recent application studies confirm the family's suitability for waste: improved YOLOv8-based detectors reach mean Average Precision in the high-eighties on curated garbage sets [14, 15], and machine-vision systems built around such detectors have been deployed for automated garbage sorting [16]. This thesis adopts YOLO11 as a single-class localizer so that the downstream classifier operates on a clean crop rather than a cluttered scene.

2.4 Public Datasets for Waste Recognition

Progress in the field is shaped by the available datasets, which differ markedly in scale, label granularity, and realism (Table 2.1). TrashNet [8] provides clean studio photographs and is the de-facto classification baseline, but its six classes and controlled conditions limit its realism. The twelve-class Garbage Classification set [9] is larger and more varied. For detection in context, TACO [17] annotates litter "in the wild" across beaches and streets, ZeroWaste [18] targets deformable objects in heavily cluttered industrial streams, and WaRP [19] captures occlusion, poor lighting, and distortion at an operating recycling plant, making the latter two the closest public references to the real-world-condition setting pursued here. Notably, none of these datasets annotate handling threat or item condition, which is one motivation for the purpose-built dataset used in this work.

Table 2.1: Representative public datasets for waste recognition.

Dataset	Scale / classes	Task	Characteristics
TrashNet [8]	∼2.5k / 6	classification	clean studio images; standard baseline
Garbage-12 [9]	∼15k / 12	classification	larger, more varied
TACO [17]	∼1.5k / 60	detection	litter in context (streets, beaches)
ZeroWaste [18]	>1.8k frames	segmentation	deformable objects, heavy clutter
WaRP [19]	>10k / 28	detection, classification, segmentation	occlusion, poor lighting, distortion

2.5 Two-Stage and Multi-Signal Systems

A recurring and effective architecture is to decouple detection from classification: a detector localizes the object, and a separate, possibly more powerful, classifier categorizes the crop. This isolates the classifier from background clutter and lets each stage be optimized independently, precisely the design adopted in this thesis. Hybrid pipelines combining a YOLO detector with a downstream CNN classifier have been shown to handle complex backgrounds and varied object sizes well [15].

A complementary idea is to fuse multiple signals rather than rely on the image alone. The Multilayer Hybrid System of Chu et al. [20] combines convolutional image features with additional numerical features through a multilayer perceptron to decide recyclability, exceeding 90% on a binary recyclable-versus-other task. This is an early precedent for the multi-head philosophy of the present work, which predicts several interdependent attributes (material, condition, route, threat) from a shared representation rather than collapsing the decision into a single label.

2.6 Hazardous and Safety-Aware Waste Handling

Despite its practical importance, the safe handling of hazardous waste is markedly under-studied in the computer-vision literature. Most systems either exclude hazardous categories entirely or fold them into a flat multi-class scheme alongside ordinary materials. When an explicit hazardous category is included, performance remains challenging: prior work using an improved Mask Scoring R-CNN [21] over four categories including a hazardous class reports overall accuracy in the mid-sixties (around 65-66%). The reviewed works do not jointly model handling severity as an ordered quantity and couple it to a disposal decision. A highly accurate material classifier can still be unsafe if it mistakes a chemical container or battery for ordinary recyclable waste. Addressing this gap through ordinal threat prediction and special-handling rules is a central contribution of this thesis.

2.7 Ordinal Regression and Model Calibration

Two modelling tools underpin the threat-aware design. First, threat severity is naturally ordinal: the levels are ordered, and a model should be penalized more for confusing "low" with "high" than "low" with "moderate." Treating such a target as ordinary multi-class classification discards this ordering. CORAL [22] and its successor CORN [23] provide neural ordinal-regression formulations whose predicted ranks are guaranteed to be consistent, and which are evaluated with rank-aware metrics such as the Quadratic Weighted Kappa. This thesis uses CORN for its threat head.

Second, for a system to abstain or defer responsibly, its confidence estimates must be trustworthy. Modern deep networks are known to be over-confident; Guo et al. [24] showed that simple post-hoc temperature scaling can reduce Expected Calibration Error while preserving multiclass predictions under a positive scalar temperature. Calibrated confidences support the confidence-aware reconciliation and Unsure option used in this work, and their quality is assessed separately

for each head. This calibration step is rarely included in prior waste-recognition systems, despite its relevance to decisions that may require human review.

2.8 Summary of Related Work

Table 2.2 consolidates the most relevant prior works, their approach, and their principal limitation with respect to the goals of this thesis.

Table 2.2: Summary of representative related work.

Work	**Approach**	**Limitation w.r.t. this thesis**
TrashNet baselines [8]	CNN transfer learning, 6 classes	clean studio images; no condition/threat
Garbage-12 baselines [9]	CNN transfer learning, 12 classes	curated images; material only
YOLO-based waste detectors [14, 15, 16]	single-stage / hybrid detection	material/localization only; no safety output
MHS [20]	CNN + MLP feature fusion	binary recyclable/other; no threat or condition
Hazardous multi-class (Mask Scoring R-CNN) [21]	flat 4-class incl. hazardous	threat as one flat class ($\sim$65.8%); no ordinal severity
CORAL / CORN [22, 23]	rank-consistent ordinal regression	general method; not applied to waste threat
Calibration [24]	temperature scaling	general method; rarely used in waste systems

2.9 Limitations of Existing Works

Across the reviewed literature, five recurring limitations define the research gap. First, most results are obtained on controlled images, leaving performance under soiling, breakage, occlusion, and uneven lighting insufficiently explored. Second, binary recyclability is too coarse for disposal decisions that must distinguish reuse, recycling, landfill, and special handling. Third, item condition is rarely modeled,

although intact and broken glass can require different routes. Fourth, hazards are generally represented as a flat class, without ordered severity or an explicit connection to handling decisions. Finally, confidence calibration and a review option are often absent, which is particularly consequential for scarce safety-critical classes. The proposed two-stage, four-head system addresses these gaps through joint prediction, per-head calibration, and a regulation-grounded decision layer.

2.10 Terminologies

- **Material:** the substance an item is made of (ten classes in this work).
- **Condition:** the physical state of the item, "Good" or "Poor" (soiled, broken, or degraded).
- **Route:** the disposal decision, Reuse, Recycle, Landfill, or Special (hazardous/special handling).
- **Threat level:** an ordinal severity score on the Litter Impact Index scale (0-3) [25], capturing the harm an item poses to handlers and the environment.
- **Backbone:** the shared feature-extraction network whose output feeds all four task heads.
- **Calibration:** the alignment of predicted confidence with empirical accuracy, measured by the Expected Calibration Error.
- **Hazard recall:** the fraction of ground-truth Special-route items sent to Special or referred for review; strict Special-routing recall counts only automatic Special decisions.

2.11 Conclusion

The literature shows substantial progress in waste material classification and in-context detection, driven by public datasets and increasingly capable CNN and transformer backbones. It also reveals a gap between recognizing an object's material and deciding how it should be handled under real-world conditions.

Joint modeling of material, condition, route, and ordered threat, supported by calibrated confidence and an explicit decision layer, offers a way to narrow that gap. The next chapter presents the methodology developed for this purpose.

Chapter 3

Methodology

3.1 Introduction

This chapter presents the proposed methodology in full. The system is a two-stage computer-vision pipeline: a single-class detector localizes the waste item, and the cropped object is passed to a shared-backbone convolutional network with four task heads that jointly predict material, condition, disposal route, and an ordinal threat level. The heads are fused by a regulation-grounded decision layer with a safety backstop. The chapter describes, in order, the overall system (Section 3.2), the experimental setup, the methodology workflow, the development of the dataset and every preprocessing operation applied to it, the detection stage and the detector architecture, the proposed multi-task network and its backbone, the loss functions, the training procedure, calibration, the decision layer, and the evaluation protocol. Each processing step is described together with the concrete operation that implements it, and every figure is described at the point where it is introduced.

3.2 System Overview

At inference time the system transforms one or more photographs of an item into an explainable disposal decision through the sequence shown in Figure 3.1. Each photograph is first localized by the detector and cropped to the item; the crop is colour-normalized and resized; a single backbone encodes it into a feature vector that feeds four task heads; the per-view head outputs are aggregated to the item; and a decision layer combines them into a final route, a reconciled threat level, a confidence value, and a human-readable explanation. The design includes

validation-fitted temperature scaling before the decision layer. Calibration is evaluated separately on the three-backbone ensemble; the paired end-to-end crop comparison uses unscaled outputs.

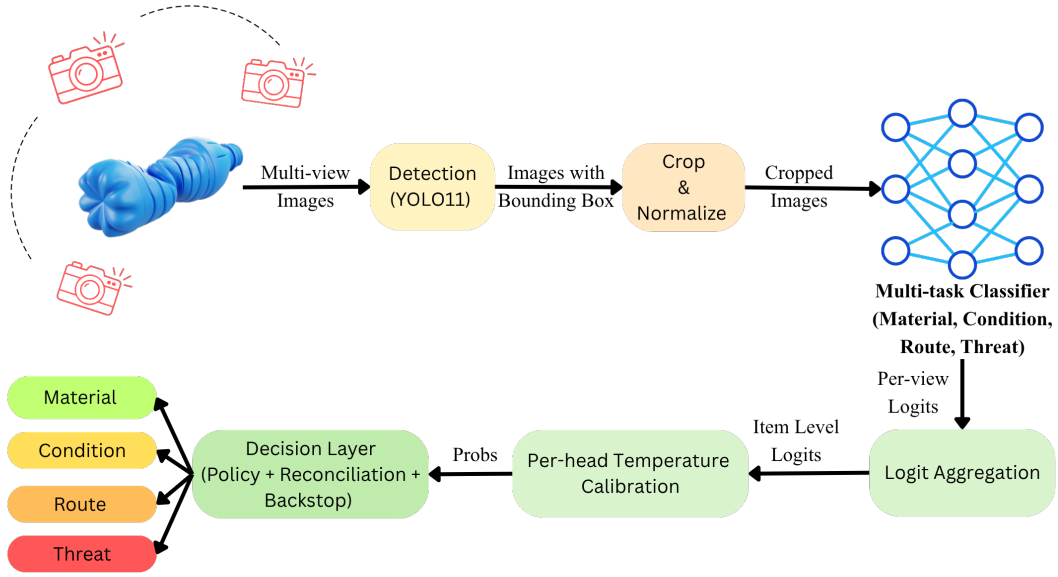


Figure 3.1: End-to-end inference pipeline of the proposed system.

Reading Figure 3.1 along its data flow: the input photograph passes through the YOLO11 detector, which returns the highest-confidence bounding box; the box is cropped and sent through the preprocessing chain (gray-world white balance, resize to 224×224, ImageNet normalization); the ConvNeXt-S backbone encodes the crop into a 768-dimensional feature that fans out to the four task heads; the per-view logits are averaged into item-level logits, with a per-head temperature-calibration stage in the design; and the decision layer finally combines them into the routing decision, threat level, confidence, and explanation. The rationale for the two-stage design is that localizing the object first reduces background clutter and focuses classification on the item. A single shared backbone feeding four lightweight heads is parameter-efficient, because the related tasks learn from one common representation.

Whereas Figure 3.1 presents the pipeline as a single linear data flow, Figure 3.2 renders the same system as a control-flow flowchart in which the decision logic and its branches are made explicit, so that the safety-critical routing behaviour can be traced from input to output.

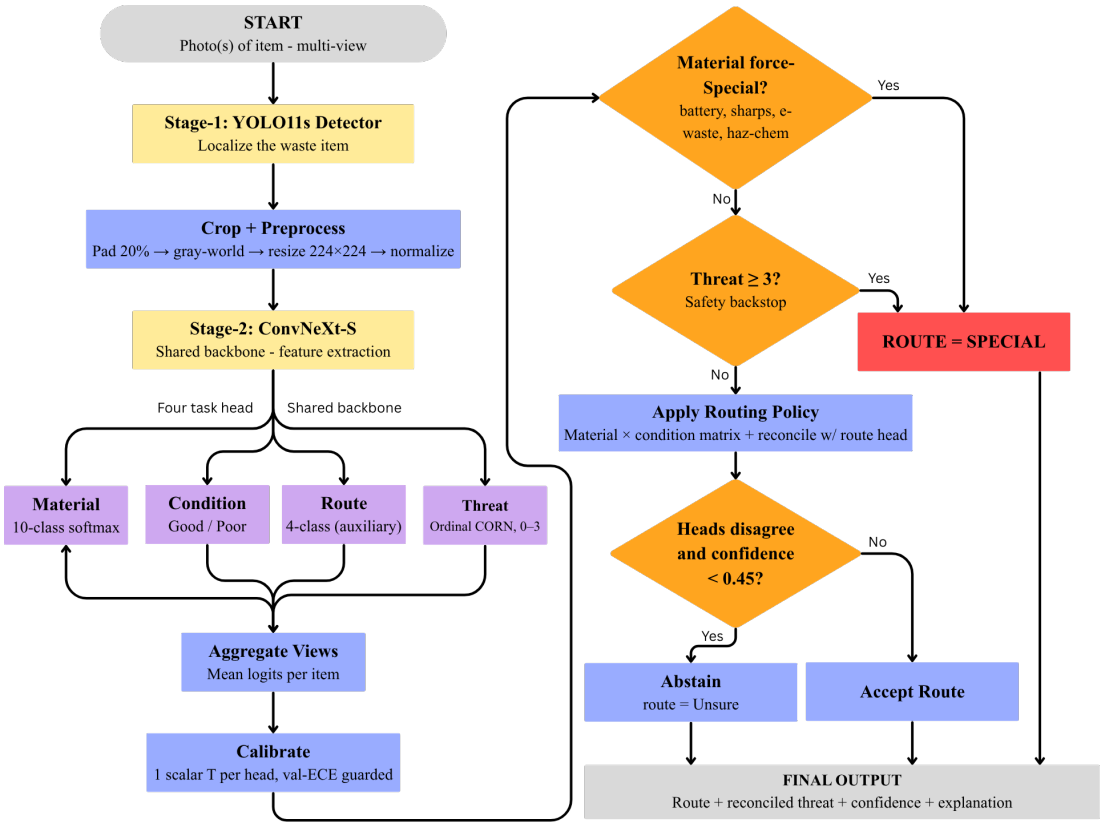


Figure 3.2: Working-process flowchart of the complete system, from the input photograph to the final routing decision, exposing the decision-layer branches and the threat-aware safety backstop.

Reading Figure 3.2 from the top left, photographs of the item enter the Stage-1 YOLO11s detector, which localizes the waste object; the crop is preprocessed (a 20% expansion of the box's longer side, gray-world white balance, resize to 224×224, and ImageNet normalization) and encoded by the Stage-2 ConvNeXt-S shared backbone, whose feature fans out to the four task heads (material, condition, route, and threat). The flowchart shows item-level averaging followed by per-head temperature calibration, and control then passes to the decision cascade in the right-hand column. That cascade is a priority-ordered sequence of tests in which the first match wins: if the predicted material is one of the force-Special streams (battery, sharps, e_waste, hazardous_chemical) the route is set to Special at once; otherwise, if the reconciled threat level is ≥ 3, the threat-aware safety backstop forces Special regardless of the material label; otherwise the material×condition routing policy is applied and reconciled with the learned route head. A final abstention gate emits "Unsure" only when the heads disagree and material confidence is below 0.45, while a Special decision is never

abstained. Every branch converges on the final route, reconciled threat, decision confidence, and explanation. The two orange decision diamonds express the system's safety constraints; the detailed rules are given in the Decision Layer section (Figure 3.13).

3.3 Experimental Setup

All experiments were implemented in Python with PyTorch, using the timm library [26] for backbone architectures and pretrained weights and the Ultralytics package [13] for the detector. Training and evaluation were performed on cloud GPU notebooks (Kaggle). A fixed random seed (42) is used for data splitting and is set for Python, NumPy, and PyTorch in the training routine to support reproducibility of augmentation and sampling. The baseline classifiers share the same losses, seed, and item split; the larger and register-token transformers use the specialised fine-tuning procedure described below. Detector candidates are compared under a common training budget and split.

3.4 Methodology Workflow

The development workflow, from raw data to a validated system, is summarized in Figure 3.3. Data are prepared once; the candidate backbones are then trained under the baseline or specialised transformer recipe; the detector is trained separately on the same items; calibration is fitted on the validation split; the decision layer is assembled; and the system is finally evaluated under several protocols.

Figure 3.3 is organized as three coloured phase bands. In the training band the item-level train split feeds two parallel lanes: the YOLO11 detector, trained with box, class, and distribution-focal losses, and the ConvNeXt-S backbone with four heads, trained with focal, binary cross-entropy, label-smoothed cross-entropy, and CORN losses under AdamW with an exponential moving average of the weights. In the validation band the per-epoch weights are evaluated: the detector by validation mAP for best-checkpoint selection, and the classifier by the composite early-stopping score followed by per-head temperature calibration. In the testing

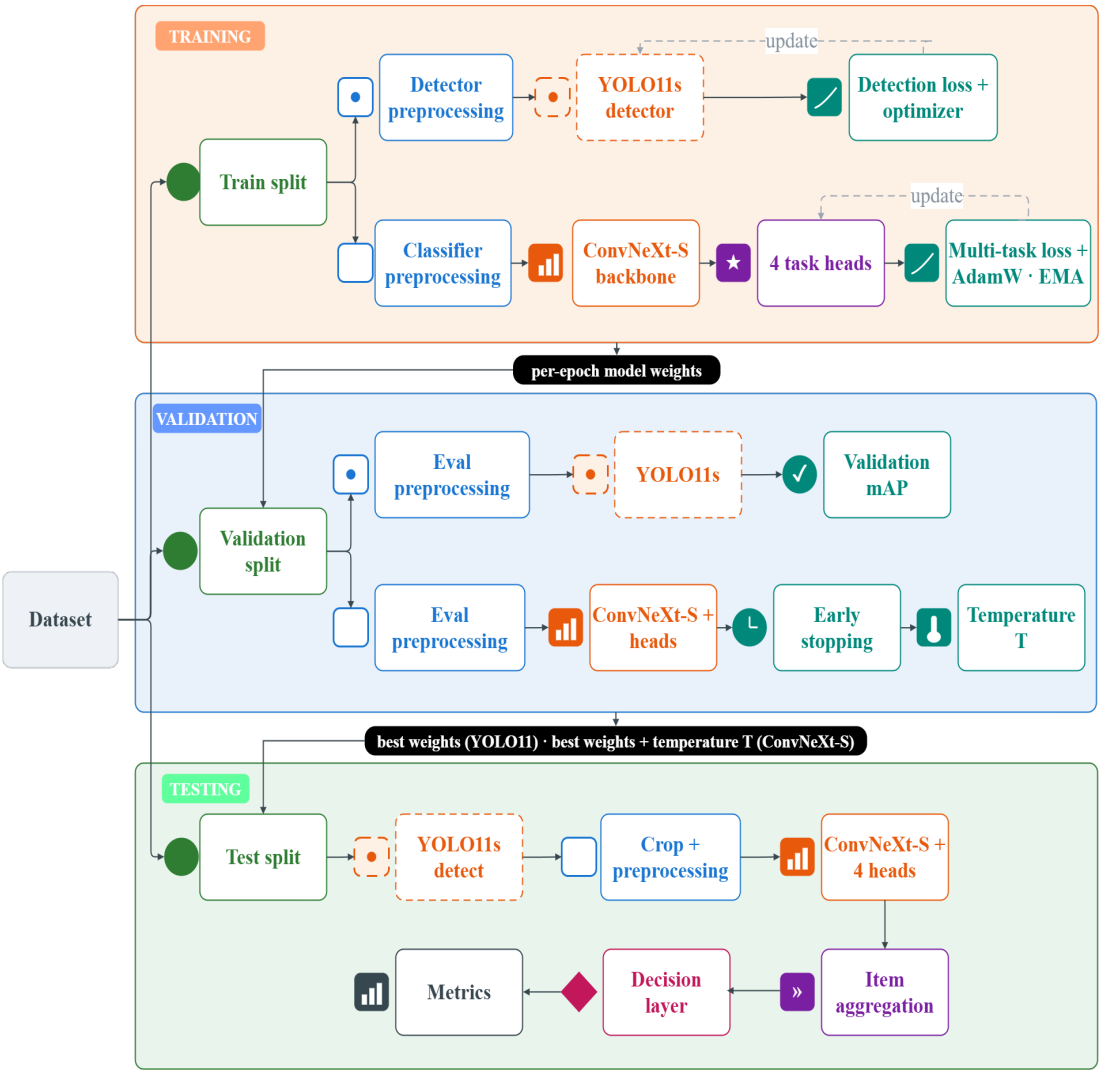


Figure 3.3: Training, validation, and testing workflow of the two-stage system.

band the models are chained end-to-end (detect, crop, classify, aggregate views, decide) and the final metrics are computed; the connectors between bands carry the per-epoch weights downward and show the best weights plus fitted temperatures entering a calibrated test. Training and validation use ground-truth classifier crops, while the end-to-end test uses predicted boxes. An additional oracle-crop test uses ground-truth boxes to quantify the localization cost separately. Model selection uses a task-weighted composite score, and calibration is fitted on validation data.

3.5 Dataset Development

The dataset was purpose-built to capture waste under varied real-world conditions and to carry the four annotations the system predicts. Figure 3.4 shows the development flow, whose substeps are described below.

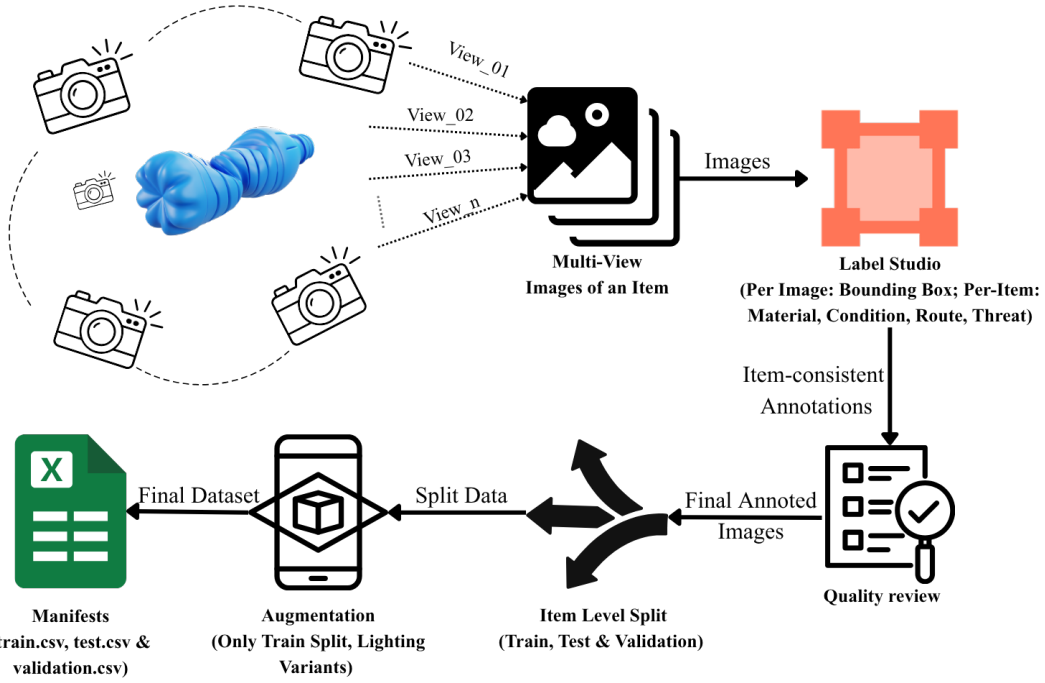


Figure 3.4: Dataset development flow from acquisition to leakage-checked splits.

Figure 3.4 traces the data from physical acquisition to the leakage-controlled manifests that the rest of the pipeline consumes: physical items are catalogued and photographed from multiple views under varied lighting and background; each item is annotated with a bounding box and the four labels; the annotated views are split by item into train, validation, and test manifests; and a leakage check confirms that no item crosses splits. Each stage of the flow is detailed in the following subsections.

3.5.1 Acquisition

The dataset comprises 485 waste items photographed under varied real-world conditions, with 5,363 image entries across the training, validation, and test manifests

(4,926, 216, and 221, respectively), including augmented training images. Lighting, background, viewing angle, and physical wear vary across the images. The physical item, not the image, is the unit of classifier and decision analysis; its several photographs are treated as different views of the same object, and classification metrics are computed after aggregating an item's views. This item-centric design mirrors the real task, in which a single decision must be made about one physical object seen from the available angles.

3.5.2 Labelling Scheme

Every item is annotated along four dimensions and with an object bounding box. The label spaces are listed in Table 3.1 and visualized in Figure 3.5.

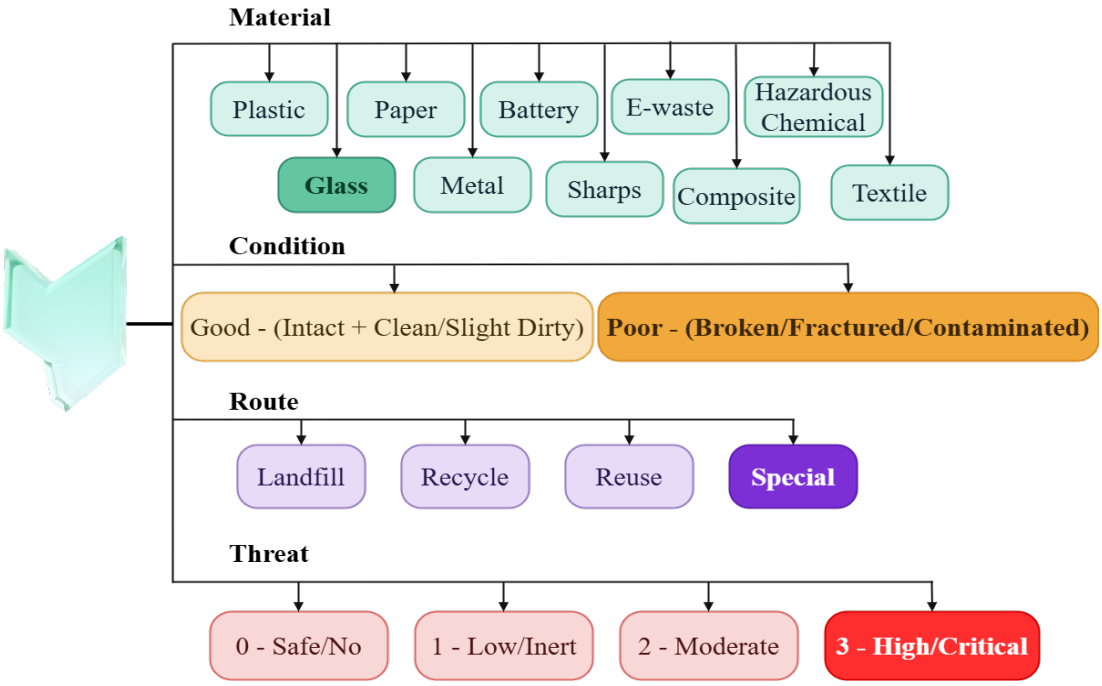


Figure 3.5: Labelling scheme: the four annotation dimensions of every item.

Figure 3.5 makes the multi-task structure explicit: a single waste item carries four simultaneous annotations, a ten-way material class, a binary condition (Good or Poor), a four-way route (Reuse, Recycle, Landfill, Special), and an ordinal threat level. Good denotes an intact item that is clean or slightly dirty, whereas Poor covers broken, fractured, or contaminated items. Threat is drawn as a graded scale to emphasize that its levels are ordered rather than nominal. The four heads share one input crop but predict distinct, partly dependent label spaces, which

is why each head is paired with a task-appropriate loss; the CORN ordinal loss exploits the threat ordering.

A concrete application of this scheme to a single real item is shown in Figure 3.6. Item 106 is a piece of broken glass photographed from several angles; one representative view carries the object bounding box tagged with the material class glass, and the two smaller views show the same physical object from other angles. All three views share a single item-level label set: material glass, condition Poor, disposal route Special, and threat level 3. The example makes the item-centric design concrete and exposes the material–condition dependency encoded by the routing policy: glass in Poor condition is escalated to Special and the maximum threat level.

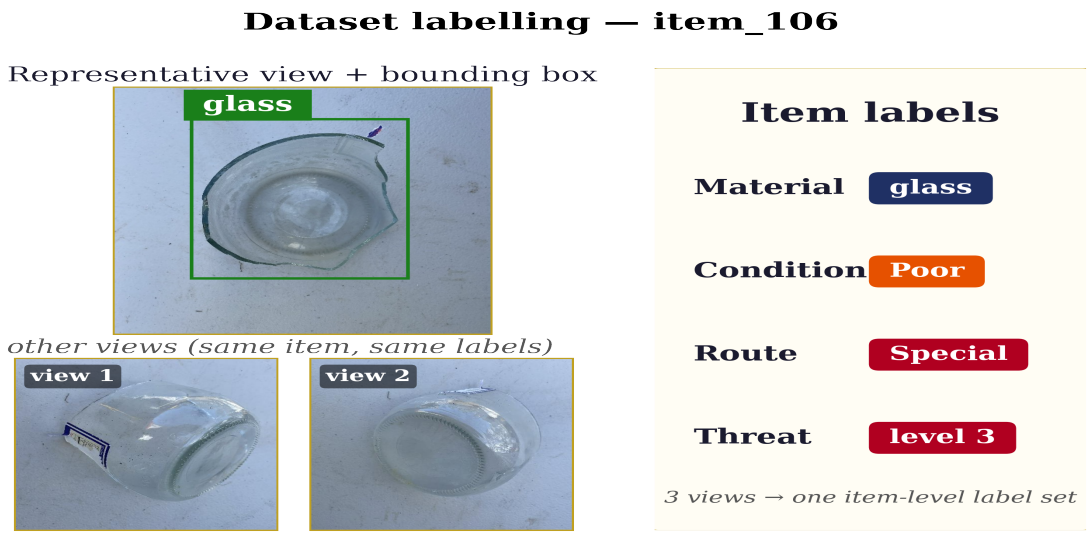


Figure 3.6: The labelling scheme applied to one real item (item 106, broken glass).

Table 3.1: Label spaces of the four prediction tasks.

Dimension	Classes
Material (10)	plastic, paper, metal, glass, textile, e_waste, battery, sharps, hazardous_chemical, composite
Condition (2)	Good, Poor
Route (4)	Reuse, Recycle, Landfill, Special
Threat (ordinal)	Litter Impact Index level 0, 1, 2, 3

Threat scale. Threat is annotated on the Litter Impact Index (LII) [25], an ordered scale from 0 (no harm) to 3 (direct human harm), defined in Table 3.2.

The dataset contains 130 level-1, 194 level-2, and 161 level-3 items, with no level-0 samples. The 0–3 range is retained for fidelity to the LII and is represented by three conditional CORN outputs; the observed threat labels occupy levels 1–3.

Table 3.2: Threat severity (Litter Impact Index) [25].

LII	Definition	Exemplars in this taxonomy
0	No harm (inert/safe)	reserved, no samples in dataset
1	Indirect harm (litter)	paper, metal, textile
2	Wildlife entanglement / ingestion	plastic, intact glass, composite
3	Direct human harm	broken glass, sharps, e_waste, battery, hazardous_chemical

Routing rules. Routing follows a Bangladesh-first regulatory basis (Solid Waste Management Rules 2021 [27]; National 3R Strategy [28]) with a hazardous "Special" safety overlay drawn from international standards (US EPA Universal Waste [29], EU WEEE [30], OSHA [31], WHO [32], CDC [33]) and Bangladesh hazardous, e-waste, and medical-waste rules [34, 35]. The deterministic routing matrix is given in Table 3.3.

Table 3.3: Routing matrix: material × condition → route.

Material	Good	Poor	Threat	Force-Special
plastic	Recycle	Landfill	2	No
paper	Recycle	Landfill	1	No
metal	Recycle	Recycle	1	No
glass	Recycle	Special	2	via condition
textile	Reuse	Recycle	1	No
e_waste	Special	Special	3	Yes
battery	Special	Special	3	Yes
sharps	Special	Special	3	Yes
hazardous_chemical	Special	Special	3	Yes
composite	Landfill	Landfill	2	if hazardous sub-part

The threat column in Table 3.3 gives the material prior; glass in Poor condition raises this prior from 2 to 3. For a predicted composite, a force-Special material among the top two material probabilities with probability above 0.15 activates the hazardous-sub-part rule.

3.5.3 Preprocessing

Each image passes through a fixed preprocessing chain before it reaches the network, implemented in the dataset's item reader (WasteDataset) and shown in Figure 3.7.

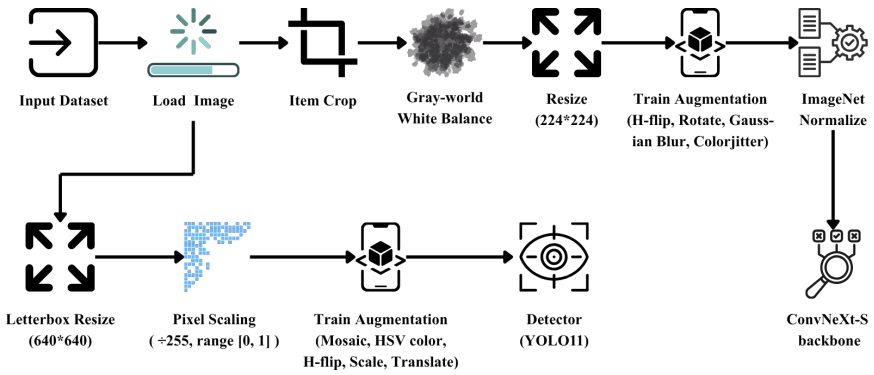


Figure 3.7: Classifier preprocessing chain.

In Figure 3.7 the raw RGB image and its bounding box enter at the top; the item is cropped with a small padding margin, gray-world white balancing suppresses colour casts introduced by varied lighting, the crop is resized to 224×224, a train-only augmentation branch applies horizontal flip, mild rotation, colour jitter, blur, and coarse dropout, and ImageNet channel normalization finally produces the tensor fed to the backbone. The chain corresponds to the following operations:

1. load_rgb reads the image as an RGB array with EXIF orientation applied.
2. object_crop crops to a square centred on the annotated box with side length $1.2 \max(w, h)$, providing a 20% expansion while retaining context. The crop is clipped to the image boundaries, and a centre square crop is used when no valid box is available.
3. gray_world applies a gray-world white balance, scaling the colour channels to a common mean to suppress the colour casts introduced by varied lighting, which is important because material appearance is colour-sensitive.
4. apply_tf resizes the crop to 224×224 and converts it to a normalized tensor using the ImageNet channel statistics; for the training split it applies the augmentation transform train_tf, and for validation and test the deterministic eval_tf.

The detector uses a separate preprocessing path (letterbox resize to 640×640 and pixel scaling to $[0, 1]$, with mosaic and colour augmentation at training time); the two paths share only the raw-image load and diverge immediately, because the detector operates on the full frame while the classifier operates on the crop.

Figure 3.8 shows both paths applied to the same real item, item 106, making the shared-input-then-diverge structure explicit. The detector path (top) letterboxes the full frame to 640×640 and scales pixels to $[0, 1]$ before the YOLO11 detector; the classifier path (bottom) crops the item with a 20% side-length expansion, applies gray-world white balance, resizes to 224×224, and applies ImageNet normalization before ConvNeXt-S. The dashed panels illustrate train-time augmentation: mosaic and colour jitter for the detector, and horizontal flip, colour jitter, and coarse dropout for the classifier. They show augmentation examples; pixel-space classifier augmentation precedes normalization and applies only to training. Validation and test use deterministic preprocessing and the evaluation-time averaging described below.

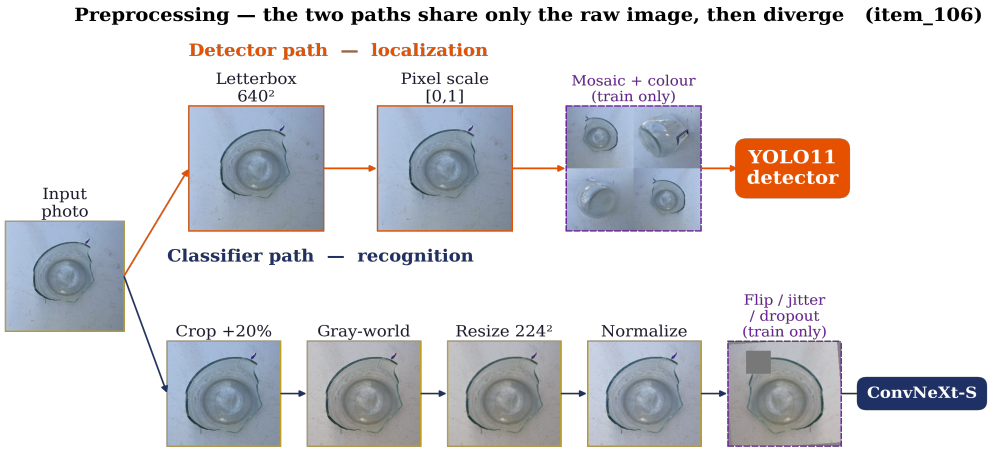


Figure 3.8: Both preprocessing paths applied to one real item (item 106).

3.5.4 Augmentation and Class Balancing

Augmentation (horizontal flips, mild rotations and colour perturbation, blur, and coarse dropout, in train_tf) is applied to the training split only; validation and test use deterministic transforms. To counter class imbalance during training, the data loader draws samples with a WeightedRandomSampler built by make_sampler, which assigns each image entry a sampling weight inversely

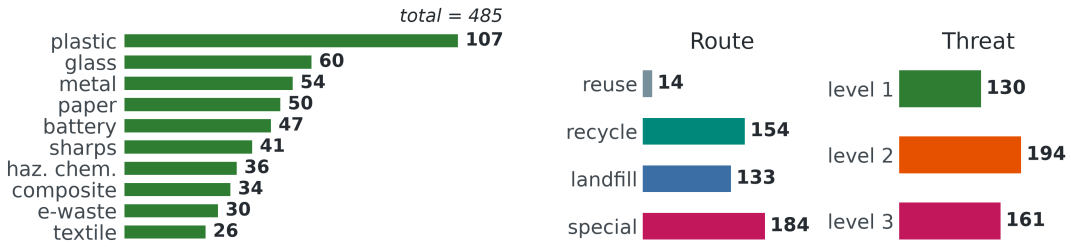
proportional to its material-class frequency and samples with replacement, so rare classes appear more often per epoch.

3.5.5 Item-Level Split and Manifests

The dataset is split by item into 385 training, 50 validation, and 50 test items, so that all views of a given item, and any augmented copies, remain within a single split. A dedicated check (check_leakage) verifies that no item identifier appears in more than one split. This item-level partitioning is essential: an image-level split would leak views of the same physical object across train and test and inflate measured performance. The split is recorded as training, validation, and test manifests that the rest of the pipeline consumes.

3.5.6 Class Distribution

The class distribution across materials, routes, and threat levels is shown in Figure 3.9; the distribution is markedly imbalanced, with plastic containing 107 items and the smallest classes, textile and e_waste, containing 26 and 30 items. Composite and hazardous_chemical contain 34 and 36 items, respectively.



(a) Ten material classes.

(b) Routes and threat levels.

Figure 3.9: Class distribution of the dataset across materials, disposal routes, and threat levels.

In Figure 3.9a the bars are sorted by frequency: common recyclables (plastic, glass, metal, paper) dominate, while textile and e_waste form the smallest classes. Figure 3.9b shows the resulting route distribution (Reuse 14, Recycle 154, Landfill 133, Special 184) and threat distribution (levels 1, 2, and 3 with 130, 194, and 161 items). Each distribution totals 485 items.

On the class imbalance. The distributions in Figures 3.9a and 3.9b reflect the

material mix and collection conditions. The route and threat distributions reflect the annotated material mix and condition: Reuse is rare because only Good textile follows the reusable route, whereas Special is common because several material classes are force-routed there. Threat level 0 is absent from the annotated sample. The imbalance is addressed at the modelling level through class-weighted focal loss and inverse-frequency sampling for the material head and class-weighted cross-entropy for the route head (Table 3.6), and at the evaluation level through macro-F1, per-class scores, and item-level cross-validation.

3.6 Detection Stage

A single-class detector is trained to localize the waste item (one "waste_item" class) from the per-item bounding-box annotations. At inference, detect_top_crop runs the detector, keeps the highest-confidence box, and crops it with crop_from_pred; if no box is returned this helper falls back to a centre crop. The paired crop evaluation instead provides a ground-truth fallback, which remained unused because every test view was detected. Localization focuses the classifier on the waste object.

3.6.1 Detector Families Considered

Two detector families are considered: single-stage YOLO detectors, which predict boxes densely in one forward pass, and transformer detectors of the DETR family, which pose detection as set prediction with attention. To choose the localizer, the members of both families summarized in Table 3.4 were compared under a common training budget (60 epochs, 640×640 input, batch 16, the same item split and seed). The empirical comparison is reported in the Results chapter.

3.6.2 YOLO11 Architecture

The selected detector is YOLO11 [13], whose internal structure is shown in Figure 3.10. It follows the canonical backbone, neck, head decomposition of modern one-stage detectors, but modernizes each part.

Table 3.4: Detector candidates and their defining architectural idea.

Detector	Family	Key architectural idea
YOLOv8	YOLO (anchor-free)	C2f backbone module and a decoupled detection head
YOLOv9	YOLO (anchor-free)	programmable gradient information with the GELAN backbone
YOLOv10	YOLO (anchor-free)	NMS-free training via consistent dual label assignment
YOLO11	YOLO (anchor-free)	C3k2 blocks, SPPF, and C2PSA partial self-attention
YOLOv12	YOLO (attention)	attention-centric backbone with area attention
RT-DETR	Transformer (DETR)	real-time transformer encoder and decoder, NMS-free

Reading Figure 3.10 top to bottom, the 640×640 input passes through Conv, C3k2, SPPF, and C2PSA backbone blocks, producing P3, P4, and P5 feature maps at strides 8, 16, and 32 (80×80, 40×40, and 20×20). C3k2 provides convolutional feature aggregation, SPPF combines pooled context, and C2PSA adds partial self-attention. The PAN/FPN neck combines features through top-down upsampling and bottom-up downsampling. At each scale, separate anchor-free head branches predict classes and box offsets, with distribution-focal training for box regression. Non-maximum suppression removes overlapping candidate detections before the highest-confidence box is selected.

3.6.3 Why YOLO11

YOLO11 was chosen for architectural and practical reasons made concrete by Figure 3.10. Its C2PSA partial self-attention and multi-scale feature fusion provide context for localizing waste under varied backgrounds; its C3k2 blocks and anchor-free decoupled head support a compact detector at the "s" scale; and the Ultralytics tooling provides a consistent path for training, validation-mAP checkpointing, and inference.

The choice is also supported by the detector benchmark in the Results chapter (Table 4.4). YOLO11s attains the highest F1 (0.9869) and precision (0.9886),

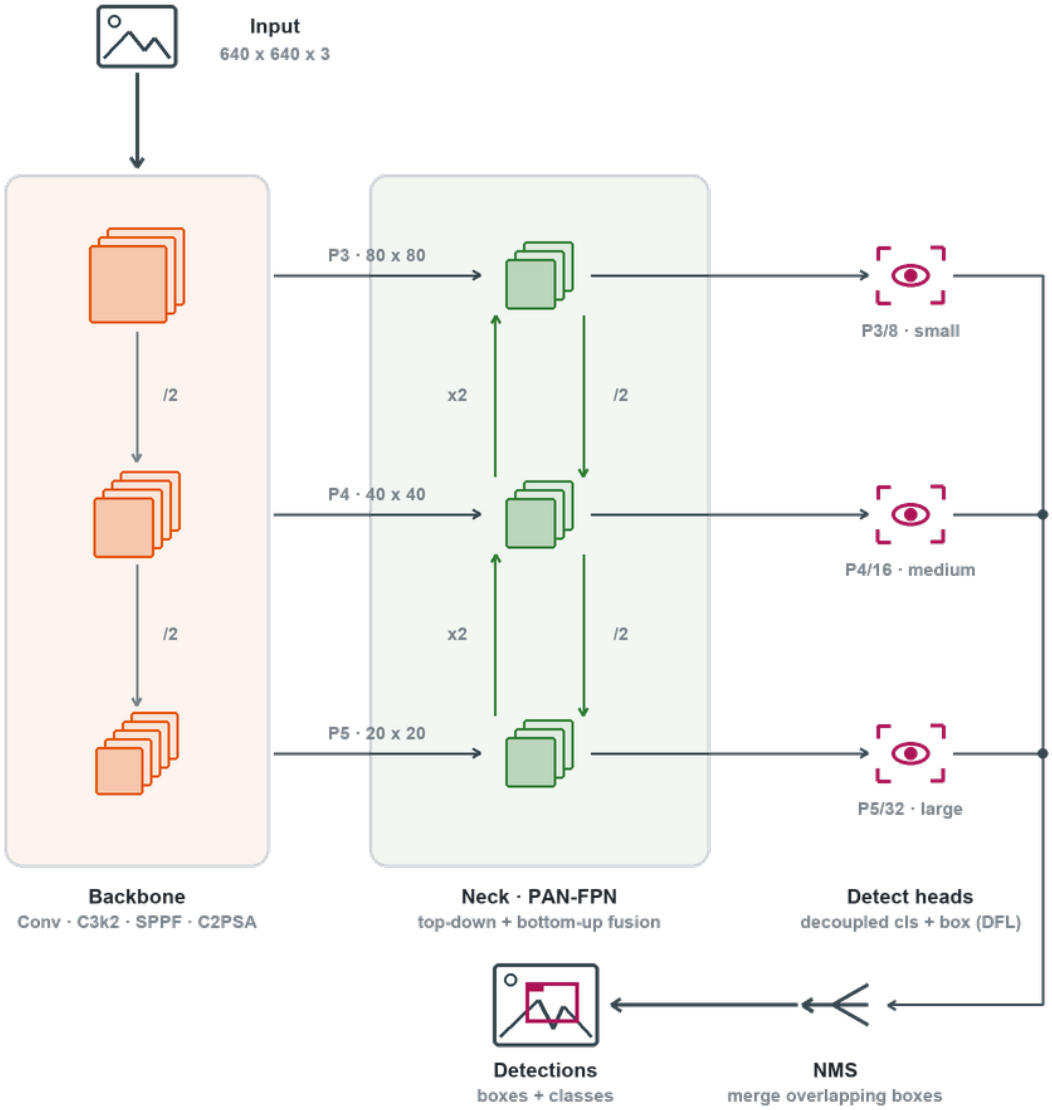


Figure 3.10: Internal architecture of the YOLO11 detector.

an mAP@50 of 0.9906, and an mAP@50:95 of 0.8458, with an inference time of 9.15 ms per image. YOLOv12s achieves higher mAP@50:95 (0.8566) at 12.81 ms, while YOLOv10s is slightly faster (9.03 ms). The choice of YOLO11s therefore balances localization accuracy and inference time. The separate evaluation of the pipeline detector on 221 test views obtains mAP@50 0.9922 and mAP@50:95 0.8534.

3.6.4 Detector Training and Inference

The detector is trained with the Ultralytics recipe (input 640×640, mosaic and colour augmentation, box, class, and distribution-focal losses) on the same training items used for the classifier, and the best checkpoint is chosen by validation mAP. At inference, detect_top_crop keeps only the highest-confidence box per image and passes the padded crop to the classifier. The detector is evaluated by mean Average Precision, and the full pipeline is additionally evaluated end-to-end against an oracle (ground-truth) crop to quantify the localization cost separately from the classification cost.

3.7 Proposed Multi-Task Network

The cropped object is encoded by one shared backbone into a feature vector that is consumed by four task-specific heads. The architecture, implemented as MultiTaskNet, uses the backbone shown in Figure 3.11.

3.7.1 Backbone Candidates

Nine backbones spanning convolutional and transformer families were evaluated as the shared trunk. Table 3.5 lists them together with their family, approximate parameter count, and pretraining. The set spans lightweight mobile CNNs (EfficientNet-B0, MobileNetV3-Large), a deep residual CNN (ResNet50), the modernized ConvNeXt family in two capacities (ConvNeXt-T, ConvNeXt-S), and four vision transformers of different scales and pretraining objectives (DINOv2 ViT-S and ViT-B, register-token DINOv2 ViT-S, and SigLIP ViT-B). The baseline recipe is given in Table 3.6; the larger and register-token transformers use the specialised small-data recipe described below. A four-backbone feature-fusion ensemble is also evaluated (Section 3.7.4).

3.7.2 ConvNeXt-S Architecture

The selected backbone is ConvNeXt-S pretrained on ImageNet-22k and subsequently fine-tuned on ImageNet-1k [5], whose structure is shown in Figure 3.11.

Table 3.5: Backbone candidates evaluated as the shared trunk.

Backbone	Family	Params (M)	Pretraining
EfficientNet-B0	CNN (mobile)	5.3	ImageNet-1k
MobileNetV3-Large	CNN (mobile)	5.5	ImageNet-1k
ResNet50	CNN (residual)	25.6	ImageNet-1k
ConvNeXt-T	CNN (modernized)	28.6	ImageNet-1k
ConvNeXt-S	CNN (modernized)	50.2	ImageNet-22k, then ImageNet-1k
DINOv2 ViT-S/14	Transformer	22.1	self-supervised
DINOv2 ViT-S/14 +reg	Transformer	22.1	self-supervised
DINOv2 ViT-B/14	Transformer	86.6	self-supervised
SigLIP ViT-B/16	Transformer	85.8	vision-language

ConvNeXt is a purely convolutional network that adopts several design choices from vision transformers while retaining the inductive bias of convolution.

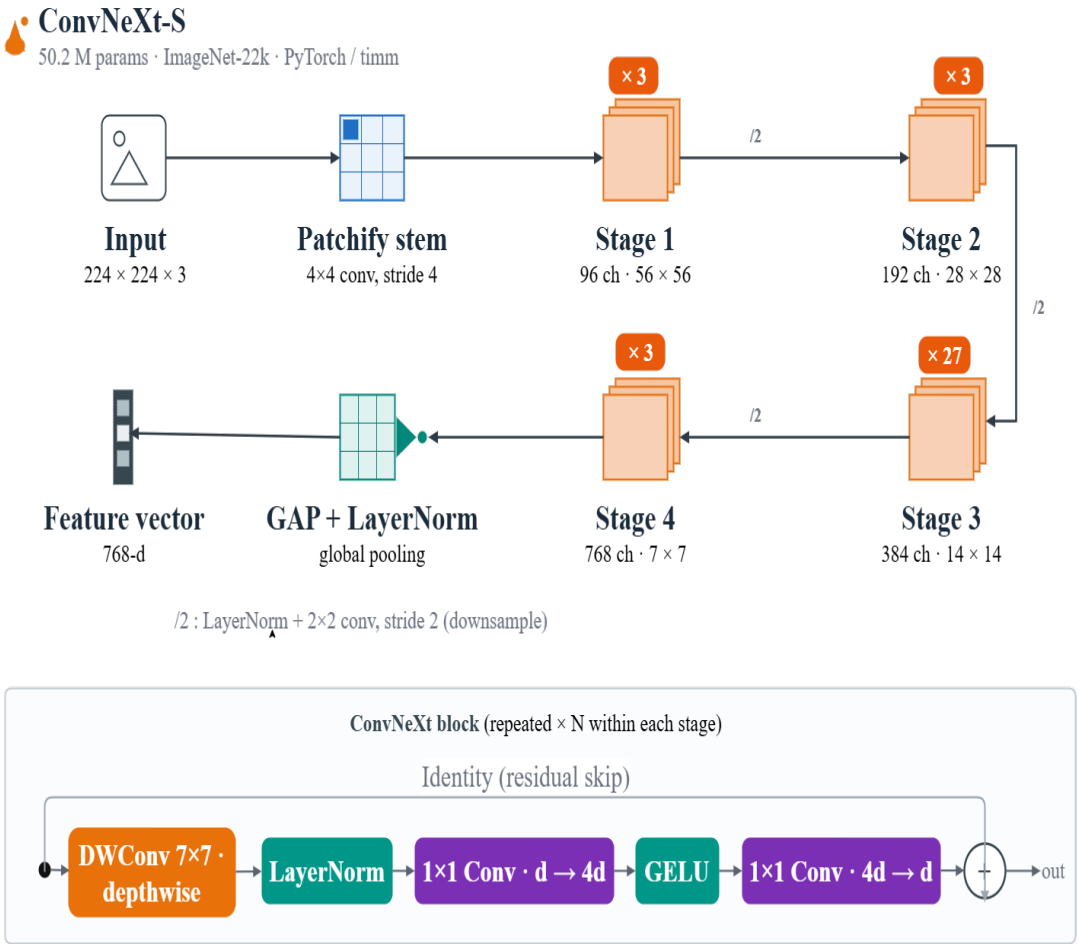


Figure 3.11: Internal architecture of the ConvNeXt-S backbone.

Figure 3.11 shows the four elements that define ConvNeXt-S. The patchify stem uses a 4×4 stride-4 convolution, analogous to transformer tokenization. The four stages follow the depth ratio $[3, 3, 27, 3]$ with channel widths $[96, 192, 384, 768]$, joined by downsampling layers (layer normalization plus a 2×2 stride-2 convolution) that halve the resolution and double the width. Each ConvNeXt block, drawn in the inset, uses a large 7×7 depthwise convolution, followed by an inverted bottleneck (layer normalization, a 1×1 expansion to four times the width, GELU, and a 1×1 projection back) and an identity residual connection. The final global average pooling and layer normalization produce the 768-dimensional feature that all four heads consume. Parameter counts in Table 3.5 are approximate architecture sizes.

3.7.3 Why ConvNeXt-S

ConvNeXt-S was selected for reasons that combine architecture and evidence. Its large-kernel depthwise convolutions provide a wide receptive field, while the deeper third stage and ImageNet pretraining supply transferable features for material discrimination. Empirically, ConvNeXt-S obtained the highest single-backbone material macro-F1 (0.8395) on the 50-item test set, ahead of DINOv2 ViT-B (0.8181). Material is central to the routing matrix and material-specific overrides. DINOv2 ViT-B performs better on route macro-F1 (0.8174 against 0.7396), threat QWK (0.8390 against 0.8261), and raw material calibration (ECE 0.0713 against 0.1102), while carrying a larger nominal architecture (86.6M against 50.2M). The selection therefore prioritizes material performance and a single shared convolutional backbone, with cross-head reconciliation and safety rules applied at the decision layer. The full per-head comparison is reported in the Results chapter (Table 4.1).

3.7.4 Vision-Transformer Fine-Tuning and Feature Fusion

The larger and register-token vision transformers are trained with a specialised small-data recipe: layer-wise learning-rate decay gives earlier blocks progressively smaller learning rates; a linear-probe-then-fine-tune schedule first warms up the task heads with the backbone frozen and then unfreezes the backbone at a low

rate; selected early blocks can remain frozen; and stochastic depth provides regularisation. Plain DINOv2 ViT-S uses the baseline recipe, so the plain and register-token variants compare both model representation and fine-tuning strategy.

Separately, a feature-fusion model concatenates the pooled features of fine-tuned ConvNeXt-S, ConvNeXt-T, ResNet50, and DINOv2 ViT-B, passes them through a small fusion network, and feeds the same four heads, with member backbones frozen so only the fusion and heads train. Fusing these domain-adapted backbones yields material macro-F1 0.8395, material ECE 0.0584, and threat QWK 0.8571. The model is reported alongside the single backbones in the Results chapter.

3.7.5 Task Heads

Each head is the same small module (Head): a linear projection to a 256-unit hidden layer, batch normalization, a GELU non-linearity, dropout at rate 0.2, and a final linear layer to the task output size. The material head outputs 10 logits, the condition head a single logit, the route head 4 logits, and the threat head $K - 1 = 3$ logits in the CORN ordinal parameterization. The forward pass returns all four outputs from one shared feature. Figure 3.12 traces the two-stage inference on item 106. YOLO11 localizes the broken glass with box confidence 0.91; the highest-confidence box is cropped to 224×224 and encoded by ConvNeXt-S, whose heads emit, after averaging the item's three views, material glass (0.99 confidence), condition Poor (P(Poor) rounded to 1.00), route Special (0.87 confidence), and threat level 3. These are the raw head predictions before the decision layer, which confirms Special through the broken-glass rule and threat backstop (Figure 3.13).

3.8 Loss Functions

The four heads are trained jointly, each with a task-appropriate loss.

Material (class-balanced focal loss). The material head uses the focal loss [36] (focal_loss), which down-weights easy examples, with per-class balancing weights α_c obtained from cb_weights (normalized inverse training-image counts

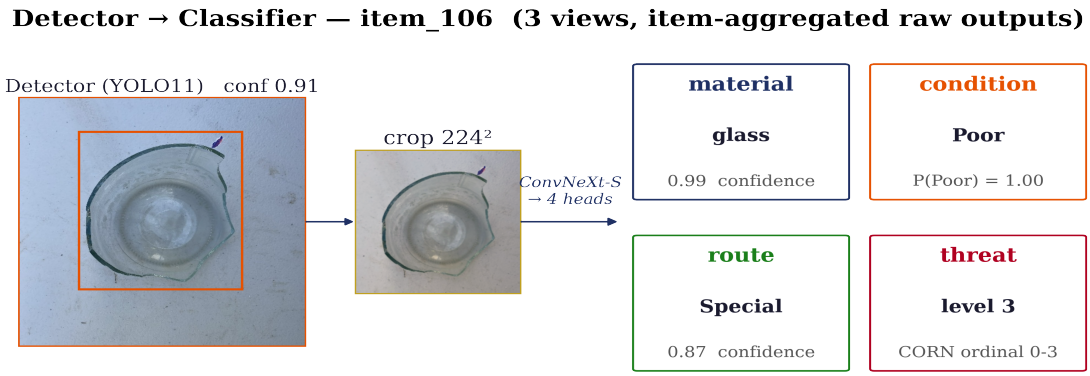


Figure 3.12: The full detect → classify inference on one real item (item 106).

$1/(n_c + 1)$):

$$\mathcal{L}_{\text{mat}} = -\,\alpha_c\,(1 - p_t)^{\gamma}\,\log(p_t), \tag{3.1}$$

where p_t is the predicted probability of the true class and $\gamma = 2$ is the focusing parameter.

Condition (binary cross-entropy).

$$\mathcal{L}_{\text{cond}} = -\Big[y \log \sigma(z) + (1 - y) \log(1 - \sigma(z))\Big]. \tag{3.2}$$

Route (weighted cross-entropy with label smoothing). The route head uses cross-entropy with class weights w_r (from cb_weights, to counter the dominance of Special and scarcity of Reuse) and label smoothing:

$$\mathcal{L}_{\text{route}} = -\sum_{k} w_k\,\tilde{y}_k \log p_k. \tag{3.3}$$

Threat (CORN ordinal loss). Because threat is ordered, the threat head uses the rank-consistent CORN formulation [23] (corn_loss), which decomposes the $K = 4$-level ordinal task into $K - 1 = 3$ conditional binary problems. Products of the conditional sigmoid probabilities produce non-increasing cumulative probabilities. Head metrics decode the level by counting cumulative probabilities above 0.5, whereas routing uses the most probable threat class.

Aggregation. The total objective is a weighted sum of the four losses with fixed weights LOSS_W: material 1.0, condition 0.5, route 1.0, and threat 1.0. An optional uncertainty-based weighting is available but left off in these experiments.

3.9 Training Procedure

The baseline backbones are trained by a single routine (train_backbone → _fit) under the recipe in Table 3.6. The procedure is:

1. **Parameter groups.** Parameters are split into a backbone group and a head group, optimized by AdamW with a small backbone learning rate (3×10^{-5}) and a larger head rate (3×10^{-4}), so the pretrained features are preserved while the new heads adapt quickly.

2. **Schedule.** A cosine learning-rate schedule with a two-epoch linear warmup is applied per step.

3. **Mixed precision.** Forward and backward passes run under automatic mixed precision (GradScaler); gradients are clipped to a norm of 1.0 for stability.

4. **Weight averaging.** An exponential moving average of the weights (ModelEmaV2, decay warmed up to 0.998) is maintained and used for validation and checkpoint selection.

5. **Model selection.** After each epoch the EMA model is evaluated on the validation split, and a composite score, $0.4\,(\text{route F1}) + 0.4\,(\text{threat QWK}) + 0.2\,(\text{material F1})$, is computed; the best-scoring checkpoint is retained.

6. **Early stopping.** Training stops if the validation score does not improve for six consecutive epochs (a maximum of 24).

7. **Test-time augmentation.** At evaluation, logits are averaged over the image and its horizontal flip before item aggregation. A separate augmentation experiment averages four deterministic variants: the base image, horizontal flip, zoom, and zoom with horizontal flip.

The specialised transformer procedure retains these losses, EMA, and validation score while using probing and layer-wise fine-tuning as described in Section 3.7.4.

Table 3.6: Training hyperparameters.

Hyperparameter	**Value**
Input resolution	224×224
Batch size	32
Optimizer	AdamW
Learning rate (heads / backbone)	3×10^{-4} / 3×10^{-5}
Schedule	cosine with 2-epoch warmup
Max epochs	24 (early stopping, patience 6)
Weight decay	1×10^{-3}
EMA decay	0.998
Label smoothing (route)	0.05
Class imbalance	class-balanced focal + inverse-frequency sampling
Random seed	42

3.10 Calibration

Modern networks can be over-confident, so temperature scaling is evaluated after training [24]. A positive scalar T fitted on the validation split rescales the material and route logits, $\hat{p} = \text{softmax}(z/T)$, while condition uses $\sigma(z/T)$ and threat uses the corresponding CORN probabilities. Separate fitters are used for the head types: fit_T_softmax (material and route), fit_T_binary (condition), and fit_T_corn (threat). A positive scalar preserves the softmax arg-max and the condition threshold at 0.5; however, temperature scaling can change the cumulative CORN prediction, so threat metrics are recomputed after calibration. The guarded scalar procedure in Figure 3.2 retains a fitted temperature only when it improves validation ECE, otherwise using $T = 1$.

The calibration experiment uses a three-backbone fusion of ConvNeXt-S, ResNet50, and DINOv2 ViT-S. Material, route, and condition temperatures are 1.0000, 0.9965, and 3.5193, respectively; the threat head uses separate cut-point temperatures $[0.5000, 2.6435, 3.3070]$, rather than the shared scalar illustrated in the flowchart. Test ECE changes from 0.1143 to 0.1143 for material, 0.1894 to 0.1888 for route, 0.2191 to 0.1120 for condition, and 0.0871 to 0.1015 for

threat. Calibration therefore substantially improves condition confidence, while the threat result illustrates the need to assess calibration separately for each head. Calibration quality is reported with ECE and reliability diagrams. Calibrated confidences can support the abstention gate, while the paired end-to-end evaluation uses unscaled probabilities.

3.11 Decision Layer

The four head outputs are fused by the confidence-aware decide engine shown in Figure 3.13. The earlier pooled policy evaluation uses apply_policy, which omits learned-route and threat-prior reconciliation and refers remaining low-material-confidence items for review. The diagrammed engine is used for the paired end-to-end evaluation.

Following the stages of Figure 3.13, the four item-level head outputs enter at the top. Stage 1 reconciles threat by taking the maximum of the material-and-condition prior and the most probable threat-head class; glass in Poor condition raises the prior to 3. Stage 2 applies the priority-ordered policy rules: force-Special materials, the threat backstop, glass in Poor condition, and a composite with a hazardous material among its top two predictions (probability > 0.15), followed by the material×condition matrix and Reuse gate. Reuse is allowed only if condition is Good and threat ≤ 1; a reconciled threat ≥ 3 forces Special independently of the material label. Stage 3 reconciles the policy route with the learned route head: Special from either estimate is retained, agreement preserves the route, and a disagreeing learned route is accepted only if it is more conservative and its confidence is at least 0.70. Stage 4 abstains ("Unsure") only when a route or threat conflict accompanies material confidence below 0.45, and Special is never abstained. The outputs are the final route, reconciled threat, a confidence value (the minimum of material, threat, and route-head confidences), and a rule-trace explanation. Decision quality is summarized by hazard-weighted cost, coverage, and hazard recall. Hazard-handling recall counts ground-truth Special items either routed to Special or referred as Unsure; strict Special-routing recall

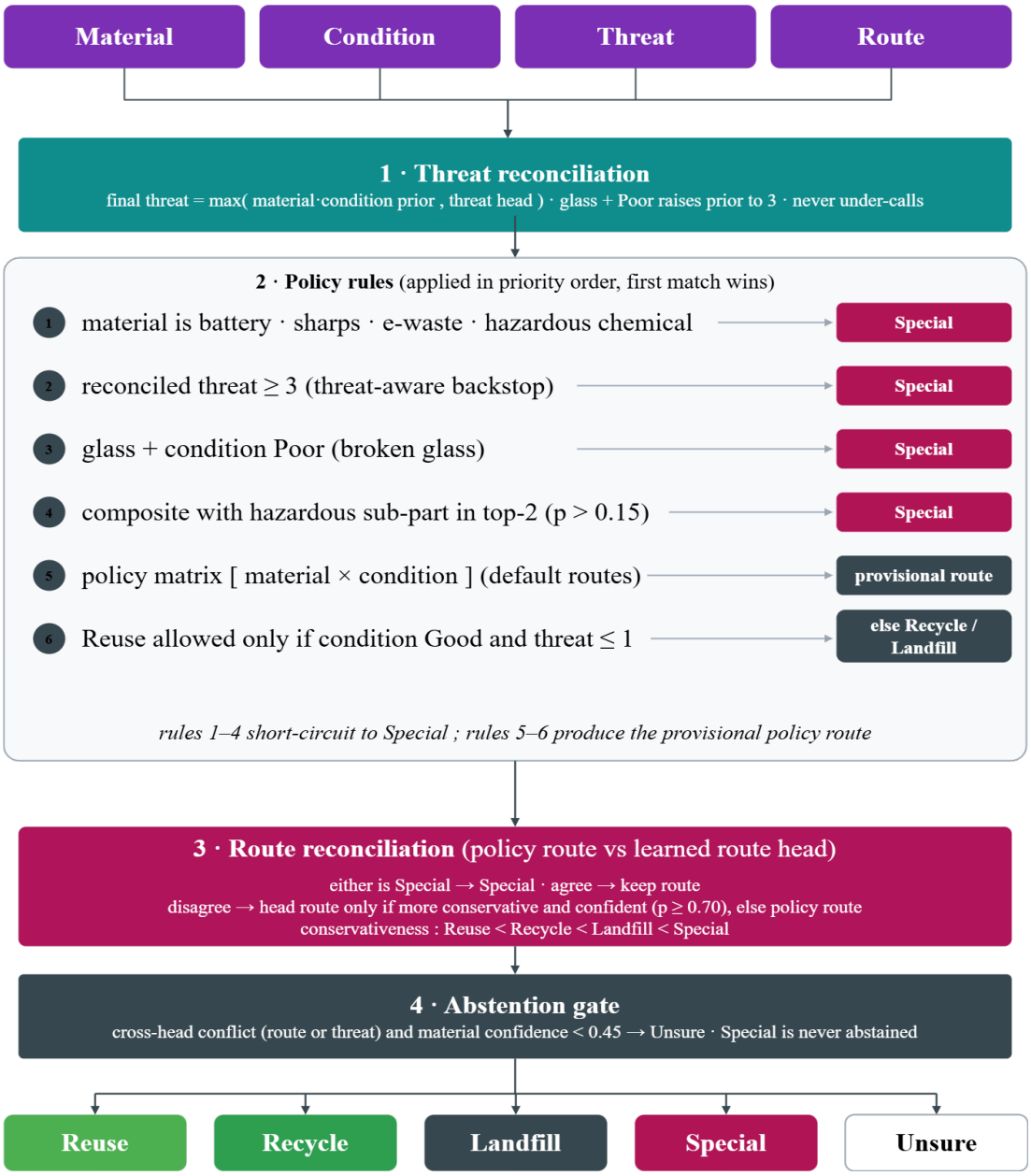


Figure 3.13: Stages of the decision layer.

counts only those routed to Special. Abstentions are excluded from accumulated cost and reported through coverage.

3.12 Evaluation Protocol

Predictions are produced and scored at the item level: evaluate_logits collects per-image head logits, aggregate_by_item averages the logits of an item's views, and item_metrics/full_report compute per-head metrics (material accuracy and

macro-F1, route macro-F1, condition AUROC, threat QWK and MAE, and per-class scores). Two protocols are used: a single held-out split of 50 items with 221 test views, and pooled out-of-fold evaluation covering 457 items and 1,961 views. Out-of-fold splitting is performed over items, stratified by material, with an internal validation subset for checkpoint selection. The 457-item evaluation precedes the expansion to the 485-item dataset described above. Bootstrap 95% confidence intervals are computed from 5,000 item resamples. The detector is scored by image-level precision, recall, and mean Average Precision; oracle and detector crops are compared to measure the end-to-end localization effect. External generalization is assessed on TrashNet and TACO using the applicable material mappings and detection annotations.

3.13 Conclusion

This chapter detailed the complete methodology: the purpose-built real-world dataset and its leakage-controlled item-level split, the per-image preprocessing chain, the two-stage detect-then-classify architecture, the choice and internal structure of the YOLO11 detector and the ConvNeXt-S backbone, the four task heads, the loss functions, the training procedure, per-head calibration, the regulation-grounded decision layer with its safety backstop, and the item-level evaluation protocol. Each step was tied to the concrete operation that implements it, and each figure was described at the point where it was introduced. The next chapter evaluates the resulting system.

Chapter 4

Results and Discussions

4.1 Introduction

This chapter evaluates the proposed system. It first defines the performance metrics used for the four heads, the detector, and the decision layer, then presents the classification backbone benchmark, the headline results with confidence intervals, calibration, the detector benchmark, detection and end-to-end performance, external validation, an interpretability and qualitative walkthrough, and an analysis of the hardest rare classes. Two complementary evaluations are used throughout. The deployed model, trained on the training split, is measured on the held-out 50-item test split. In addition, pooled out-of-fold predictions over an earlier 457-item dataset provide a robustness estimate with bootstrap confidence intervals.

4.2 Performance Evaluation Metrics

4.2.1 Precision, Recall, and F1-Score

Precision is the fraction of predicted positives that are correct, recall the fraction of true positives that are retrieved, and F1 their harmonic mean:

$$\text{Precision} = \frac{TP}{TP + FP}, \qquad \text{Recall} = \frac{TP}{TP + FN}, \tag{4.1}$$

$$\text{F1} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}. \tag{4.2}$$

For multi-class heads we report macro-F1 (the unweighted mean of per-class F1), which counts rare classes equally and is therefore the fair metric under imbalance.

4.2.2 mAP@50 and mAP@50-95 (detection)

Detection quality is measured by mean Average Precision (mAP). The Intersection over Union (IoU) between a predicted box and the ground truth is

$$\text{IoU} = \frac{\text{area of overlap}}{\text{area of union}}. \tag{4.3}$$

mAP@50 is the mean Average Precision at an IoU threshold of 0.50, and mAP@50-95 averages mAP over IoU thresholds from 0.50 to 0.95 in steps of 0.05.

4.2.3 Quadratic Weighted Kappa (threat)

Because threat is ordinal, it is evaluated with the Quadratic Weighted Kappa (QWK), which rewards predictions that are close on the 0-3 scale and penalizes large rank errors quadratically. A QWK of 1 is perfect agreement and 0 is chance-level.

4.2.4 AUROC, ECE, and Hazard Recall

The binary condition head is evaluated with the threshold-free AUROC. Calibration is measured with the Expected Calibration Error (ECE), the average gap between confidence and accuracy across 15 confidence bins. The decision layer is judged primarily by hazard recall: the fraction of truly hazardous (Special-route) items routed to Special or referred for human review. Strict Special-routing recall counts only automatic Special decisions, and coverage measures the fraction of all items receiving an automatic decision. Reporting these together distinguishes safe referral from successful automatic routing.

4.3 Classification Backbone Benchmark

Ten classifiers spanning three architectural families were evaluated on the same held-out split at the item level: five CNNs, four self-supervised or vision-language vision transformers (DINOv2 ViT-S, DINOv2 ViT-B, DINOv2 ViT-S with register tokens, and SigLIP ViT-B), and a four-backbone feature-fusion ensemble.

The candidates share the task definitions and data split, with transformer-specific fine-tuning as described in the methodology chapter. Table 4.1 and Figure 4.1 summarize their performance. ConvNeXt-S (ImageNet-22k) is the strongest single backbone on the primary material task and is used in the deployed pipeline. The final row of Table 4.1 is a best-per-head *oracle*: it combines the best value in each column, with material accuracy and macro-F1 from ConvNeXt-S or fusion, route from ResNet50, threat QWK and material ECE from fusion, and condition AUROC from DINOv2 ViT-S. This row indicates the performance available across several models rather than the outcome of one shared-backbone network.

Table 4.1: Per-backbone results on the held-out split (item-level); best per column in bold.

		Material		**Route**	**Threat**	**Cond.**	**Mat.**
Backbone	**Family**	**acc**	**F1**	**F1**	**QWK**	**AUROC**	**ECE**
ConvNeXt-S (IN-22k)	CNN	**0.840**	**0.840**	0.740	0.826	0.865	0.110
Fusion (4-bb)	Ensemble	**0.840**	**0.840**	0.740	**0.857**	0.874	**0.058**
ConvNeXt-T	CNN	0.820	0.823	0.750	0.840	0.794	0.116
DINOv2 ViT-B	Transformer	0.820	0.818	0.817	0.839	0.890	0.071
SigLIP ViT-B	Transformer	0.820	0.814	0.768	0.775	0.833	0.123
DINOv2 ViT-S+reg	Transformer	0.760	0.770	0.666	0.840	0.850	0.201
DINOv2 ViT-S	Transformer	0.760	0.769	0.734	0.825	**0.905**	0.137
EfficientNet-B0	CNN	0.780	0.768	0.766	0.716	0.874	0.153
MobileNetV3-L	CNN	0.720	0.721	0.683	0.806	0.871	0.200
ResNet50	CNN	0.720	0.715	**0.876**	0.764	0.874	0.167
Best-per-head oracle	—	0.840	0.840	0.876	0.857	0.905	0.058

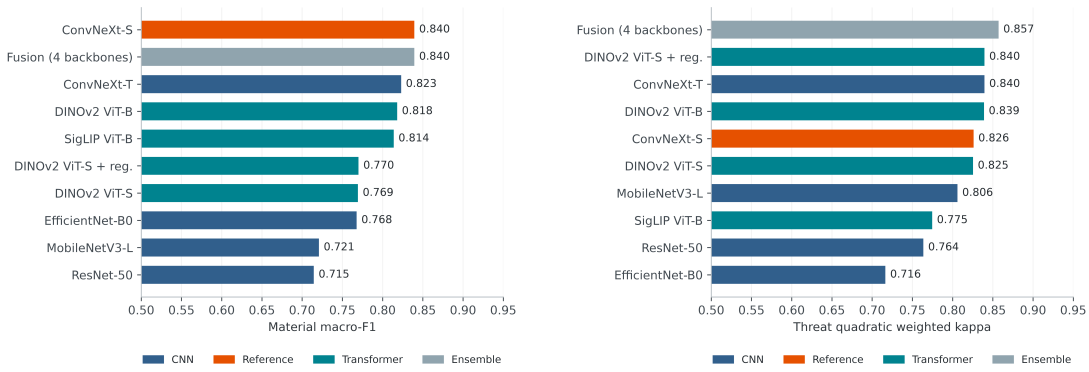
The benchmark supports four observations. First, ConvNeXt-S leads the single backbones on the primary material task, with accuracy 0.840 and macro-F1 0.840, tying the fusion ensemble on both measures. Second, the larger DINOv2 ViT-B closes much of the material gap (F1 0.818) and is the best-calibrated single backbone (ECE 0.071); the vision-language SigLIP ViT-B reaches F1 0.814, while plain DINOv2 ViT-S and its register-token variant reach 0.769 and 0.770. Third, fusion provides complementary gains: combining ConvNeXt-S, ConvNeXt-T, ResNet50, and DINOv2 ViT-B attains the best threat QWK (0.857) and the lowest material ECE (0.058), while matching the best material performance. Its threat advantage over ConvNeXt-S is 0.031, whereas the route gain is only 0.0003

and the condition AUROC gain is 0.0097. Fourth, the per-head winners remain spread across models: route favors ResNet50, threat favors fusion, and condition favors DINOv2 ViT-S. ConvNeXt-S therefore provides a material-focused shared-trunk choice, while the ensemble quantifies the headroom of a heavier combination. Figure 4.1 shows these rankings graphically for each head.

The accuracy gains of fusion must also be weighed against model size (Figure 4.2). ConvNeXt-S contains approximately 50.25 million learned-state elements, compared with 189.15 million for fusion, a 3.76× increase. The single trunk thus provides the same material performance with a substantially smaller model. Classifier inference latency remains to be measured under a common hardware and inference setting.

These rankings are examined with paired statistics on the shared test items: a paired bootstrap of the material macro-F1 difference with 10,000 item resamples and an exact McNemar test on material correctness. Against ResNet50, ConvNeXt-S has a macro-F1 difference of +0.125 with a 95% confidence interval of [0.023, 0.259]; against DINOv2 ViT-S, the difference is +0.070 with interval [0.013, 0.141]. The corresponding McNemar p-values are 0.070 and 0.125. Macro-F1 and paired correctness assess different aspects of performance, so the bootstrap result does not imply significance under both tests. Fusion and ConvNeXt-S make identical material predictions on these items. With only 50 test items and no multiple-comparison adjustment, selection rests on the primary-task point estimates and the simpler single-trunk design rather than a claim of universal statistical superiority.

A final control examines whether four-head sharing is bought at the primary task's expense. The same ConvNeXt-S backbone was retrained with a material-only objective, with the route, condition, and threat losses switched off. It reaches the same item-level material accuracy and macro-F1 (0.840 and 0.840) as the four-head model. Sharing one backbone therefore adds the three additional outputs without a measurable material-score penalty on this split, supporting the practical value of the multi-task design.



(a) Material macro-F1

(b) Threat QWK

(c) Route macro-F1

(d) Condition AUROC

Figure 4.1: Backbone comparison per head.

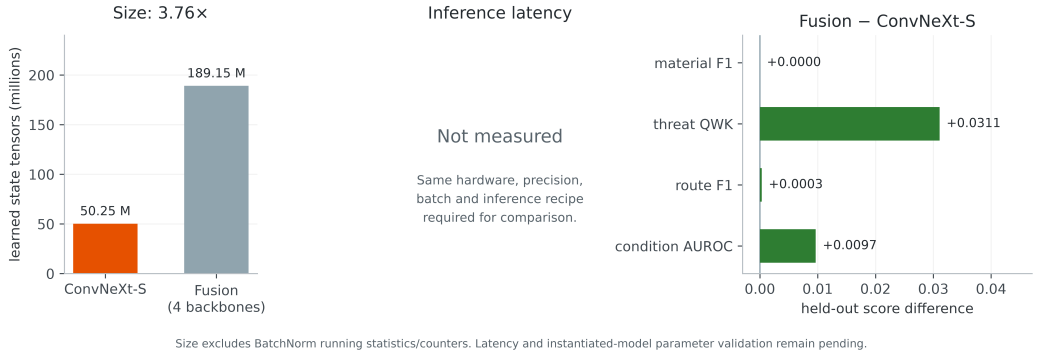


Figure 4.2: Deployed ConvNeXt-S versus the four-backbone fusion: learned-state size and per-head performance differences. Fusion gains 0.031 threat QWK with 3.76× the learned-state elements; inference latency remains unmeasured.

4.4 Headline Results: Deployed Model and Cross-Validated Robustness

The deployed model is evaluated on the held-out 50-item test split, and its robustness is examined through pooled out-of-fold predictions for 457 items from the earlier dataset. Bootstrap 95% confidence intervals are computed with 5,000

item resamples. The two evaluations answer complementary questions: the test split measures the selected model, while the out-of-fold evaluation provides a broader view of performance across items. Table 4.2 reports both, and Figure 4.3 overlays the deployed model's scores on the cross-validated confidence intervals.

Table 4.2: Headline metrics of ConvNeXt-S and the cross-validated robustness check. Classifier test metrics use oracle crops; decision test metrics use the integrated pipeline.

Metric	**Deployed (test, n=50)**	**OOF (n=457) [95% CI]**
Material accuracy	0.840	0.832 [0.797, 0.864]
Material macro-F1	0.840	0.803 [0.760, 0.841]
Material ECE	0.110	0.036
Condition AUROC	0.865	0.821 [0.782, 0.857]
Route macro-F1	0.740	0.675 [0.608, 0.735]
Threat QWK	0.826	0.771 [0.708, 0.831]
Hazard recall (Special or review)	0.824 (14/17)	**0.888 [0.838, 0.933]** (150/169)
Strict Special-routing recall	0.824 (14/17)	0.876 (148/169)
Automatic-decision coverage	1.000	0.961 (439/457)

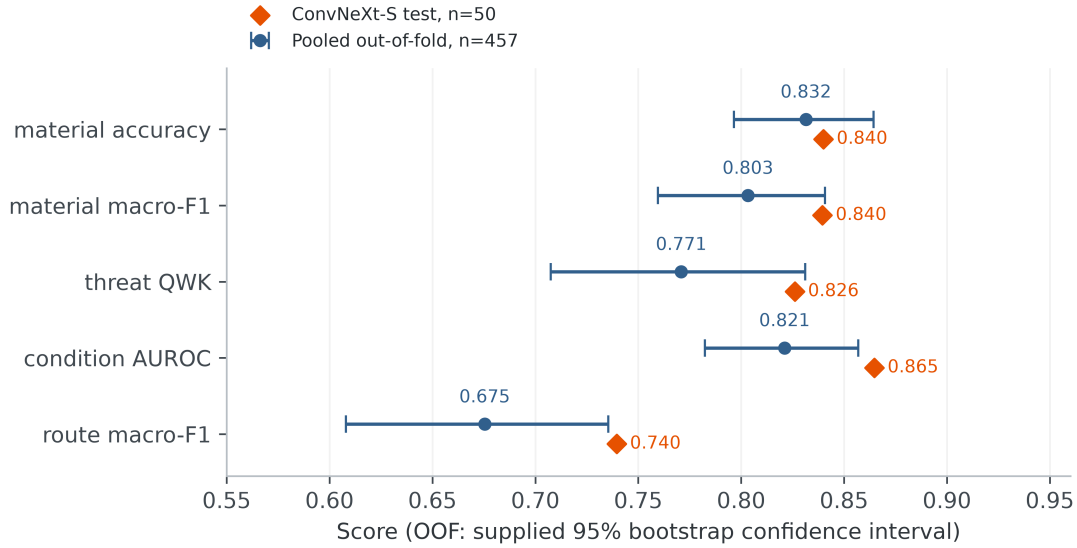


Figure 4.3: Cross-validated metrics (457-item out-of-fold pool, 95% confidence intervals) with the deployed model's 50-item test scores overlaid.

The two evaluations show similar material and threat performance. The deployed model's material accuracy and macro-F1 (both 0.840) lie inside the cross-validated intervals, as does its threat QWK (0.826). Its condition AUROC (0.865)

and route macro-F1 (0.740) lie slightly above the respective interval endpoints. Because the two evaluations use different item pools and fitted models, these differences are interpreted as complementary estimates rather than attributed solely to training-set size. Material calibration is strong in the pooled evaluation (ECE 0.036), although condition ECE is higher at 0.211.

Most importantly, the earlier policy evaluation achieves hazard recall 0.888 [0.838, 0.933] over 169 hazardous items: 148 are routed to Special and two are referred for review. Strict Special-routing recall is 0.876, with 19 hazardous items receiving ordinary routes. Overall coverage is 0.961, since 18 of 457 items are deferred. The policy uses unscaled probabilities and the most probable threat class. Its total decision cost is 459, or 1.004 per item [0.685, 1.363]; referrals contribute no automatic-routing cost, so this result is read alongside coverage. The held-out pipeline routes 14 of 17 hazardous items to Special (recall 0.824) with no referrals. These outcomes show the benefit and remaining limits of prediction-based safety routing.

4.4.1 Per-Class Material Performance

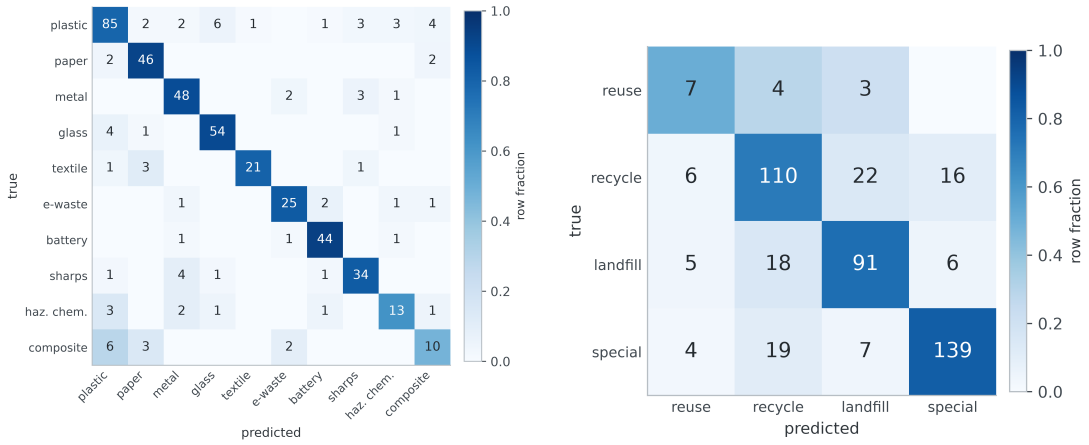
Table 4.3 reports the cross-validated per-class material F1. Eight of the ten classes exceed F1 0.80, led by battery (0.917). The weakest classes are composite (0.513) and hazardous_chemical (0.634), with 21 items each in the 457-item evaluation. These class supports belong to the earlier evaluation, while the expanded dataset contains 34 composite and 36 hazardous-chemical items. Their errors are analyzed in Section 4.9.

Table 4.3: Cross-validated per-class material F1 and support (457-item evaluation).

Class	**F1**	**n**	**Class**	**F1**	**n**
battery	0.917	47	textile	0.875	26
paper	0.876	50	glass	0.885	60
e_waste	0.833	30	metal	0.857	54
sharps	0.829	41	plastic	0.813	107
composite	0.513	21	hazardous_chemical	0.634	21

4.4.2 Confusion Matrices and ROC Curves

The cross-validated confusion matrices for all four heads are shown in Figure 4.4, and the ROC curves for condition and route in Figure 4.5. Material confusion is concentrated in composite and hazardous chemical, which are often mistaken for their visible constituent materials. Threat confusion is strongly diagonal: 382 of 457 predictions are correct, with 58 one-level and 17 two-level errors. The plot displays levels 1–3 because level 0 has no examples in this evaluation. Binary condition errors are similar in both directions (158 of 221 Good and 179 of 236 Poor items correct, accuracy 0.737). The route curves add an important nuance to its lower macro-F1: one-vs-rest AUROCs are Reuse 0.922, Recycle 0.888, Landfill 0.930, and Special 0.949. The head ranks routes well, but committing to one label under imbalance remains difficult, particularly for the 14-item Reuse class (F1 0.389) and the Recycle/Landfill boundary.



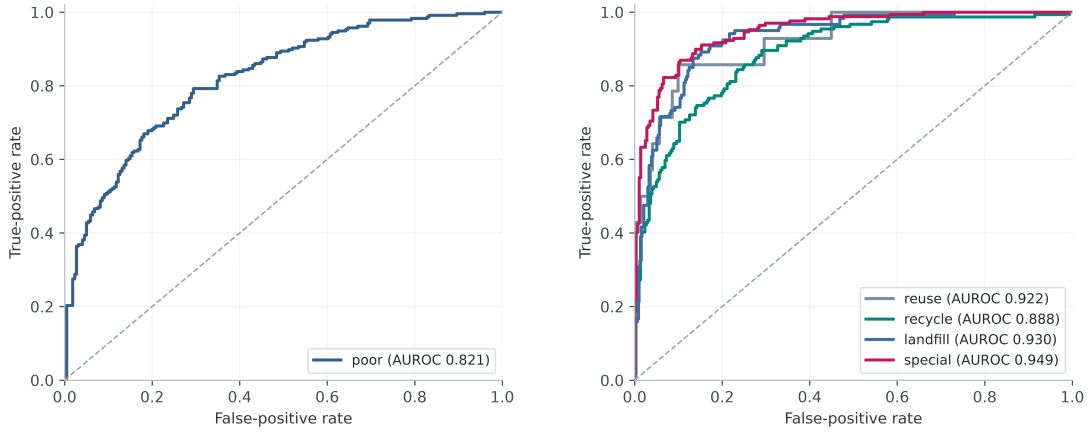
(a) Material

(b) Route

(c) Threat

(d) Condition

Figure 4.4: Cross-validated confusion matrices for the four task heads.



(a) Condition (Good vs Poor)

(b) Route (one-vs-rest)

Figure 4.5: ROC curves for the condition and route heads.

4.4.3 Calibration and Training Behaviour

Figure 4.6 (left) shows the material reliability diagram over the cross-validated pool, with ECE 0.036, matching Table 4.2. The right panel shows the effect of validation-fitted temperature scaling on the earlier three-backbone fusion. It is most effective for condition: a temperature of about 3.52 reduces test ECE from 0.219 to 0.112. Material remains at 0.114 with temperature 1, and route changes only slightly (0.189373 to 0.188798, both rounded to 0.189) with temperature about 1.00. Threat scaling uses three cut-point temperatures [0.500, 2.644, 3.307] and increases ECE from 0.087 to 0.102. Calibration therefore benefits some heads more than others, reinforcing the need to assess each head separately before its confidence is used for routing or abstention.

Figure 4.7 shows the training behavior of the selected ConvNeXt-S backbone. The left panel plots the weighted multi-task training loss, which falls from 3.213 to 0.374 and stabilizes after approximately six epochs. The right panel plots item-level validation metrics for all four heads. The checkpoint is chosen by the composite score $0.4 \cdot \text{route F1} + 0.4 \cdot \text{threat QWK} + 0.2 \cdot \text{material F1}$, which first peaks at epoch index 10 (dashed line). Later training does not consistently improve these metrics, while validation loss rises from 2.612 to 3.375. EMA weights and early stopping therefore support model selection before continued fitting increases the gap between training and validation behavior.

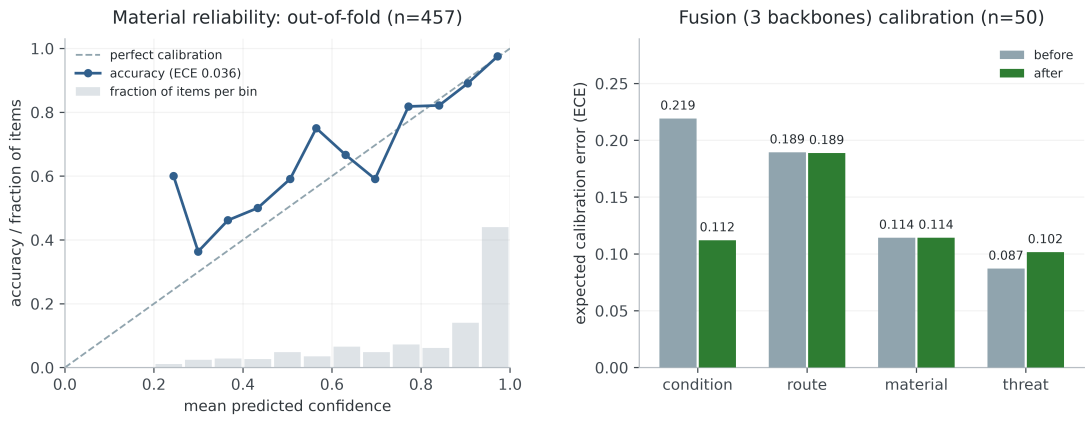


Figure 4.6: Cross-validated material reliability and per-head ECE before/after temperature scaling of the earlier three-backbone fusion.

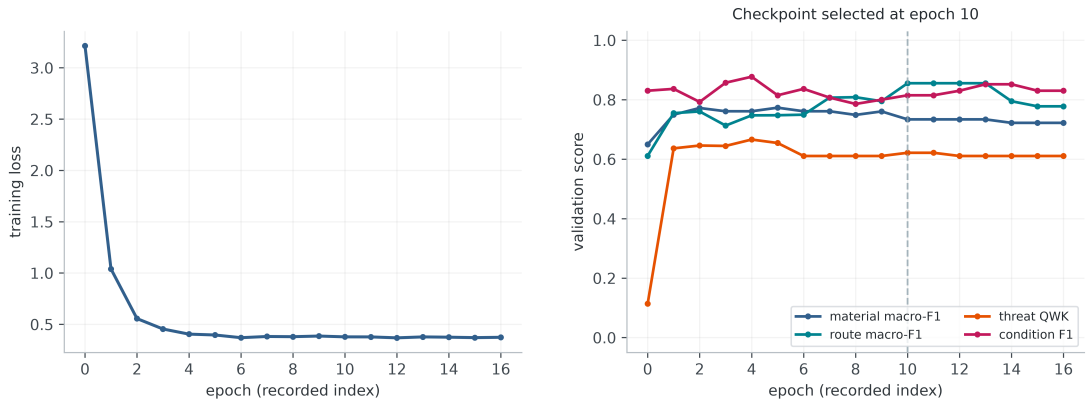


Figure 4.7: Multi-task training loss and per-head validation metrics of ConvNeXt-S.

4.5 Detection and End-to-End Performance

4.5.1 Detector Benchmark

To justify the choice of detector empirically, six detectors, five modern YOLO variants at the same small scale and the transformer-based RT-DETR-L, were trained under a common budget (60 epochs, 640×640 input, batch 16, the same item split and seed) and evaluated on the held-out test split. Table 4.4 reports accuracy, model size, and measured inference latency.

Three conclusions follow. First, all five YOLO variants achieve high mAP@50 (0.979–0.992) in the single-item setting, making strict-IoU accuracy, F1, and

Table 4.4: Detector comparison on the held-out test split; best per column in bold.

Detector	P	R	F1	mAP@50	mAP@50:95	Par. (M)	ms/img
YOLOv12-s	0.985	**0.986**	0.986	**0.992**	**0.857**	9.3	12.8
YOLOv9-s	0.956	0.968	0.962	0.979	0.849	**7.3**	10.8
YOLO11-s (ours)	**0.989**	0.985	**0.987**	0.991	0.846	9.5	9.2
YOLOv10-s	0.974	0.966	0.970	0.984	0.837	8.1	**9.0**
YOLOv8-s	0.971	0.981	0.976	0.989	0.825	11.2	9.3
RT-DETR-L (Transf.)	0.872	0.904	0.888	0.905	0.738	33.0	39.1

latency useful discriminators. Second, YOLO11-s offers a favorable joint trade-off: the highest F1 (0.987) and precision (0.989), mAP@50 0.991, and mAP@50-95 0.846, within 0.011 of YOLOv12-s. Its latency is about 29% lower than YOLOv12-s (9.2 versus 12.8 ms), while YOLOv10-s is only marginally faster (9.0 ms) with lower F1 0.970. Third, RT-DETR-L trails the YOLO candidates in this experiment, with mAP@50-95 0.738 at 39.1 ms per image. The choice of YOLO11 is therefore supported by the measured accuracy–speed balance among the tested configurations.

4.5.2 Deployed Detector and End-to-End Cost

Re-evaluated on the expanded test split (50 items, 221 views), the deployed single-class YOLO11 detector reaches mAP@50 0.992, mAP@50-95 0.853, precision 0.978, and recall 0.989. At confidence threshold 0.25 it returns a detection for every test view, giving detection rate 1.0. Figure 4.8 shows precision–recall and confidence curves from an earlier detector evaluation, with displayed mAP@50 0.995; the expanded-split metrics are reported separately above. Detection rate indicates whether a box is returned, while IoU-based recall also measures whether the box sufficiently overlaps the object.

Running the full pipeline end-to-end (detect → crop → classify) produces the same material accuracy and macro-F1 as oracle ground-truth crops (both 0.840) on the paired 50-item evaluation (Figure 4.9). This crop comparison uses base inference without temperature scaling for both inputs, whereas the ConvNeXt-S backbone benchmark averages original and horizontally flipped crops. Threat QWK is also unchanged at 0.826. The other heads show small localization costs:

route macro-F1 falls from 0.740 to 0.724, and condition AUROC from 0.866 to 0.863. The detector therefore preserves the material result in this experiment, while its effect on downstream performance remains task-dependent.

Table 4.5 separates the two models' standalone performance from the integrated pipeline. The detector and classifier are evaluated on their respective tasks, while the paired oracle-versus-detector crop comparison makes the cost of chaining them explicit.

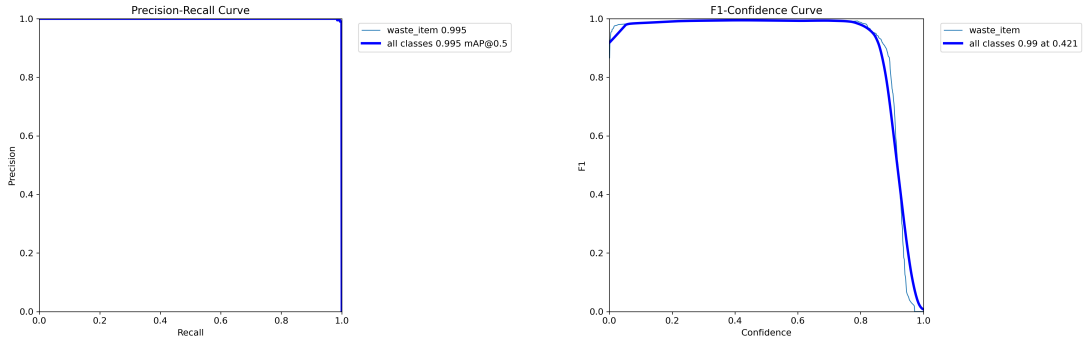
Table 4.5: Separate-model outcomes versus the integrated end-to-end pipeline.

Configuration	**Evaluation**	**Result**
YOLO11 detector (Stage 1, standalone)	test	mAP@50 0.992; mAP@50-95 0.853; precision 0.978; recall 0.989
ConvNeXt-S classifier (Stage 2, oracle crops)	pooled CV	material acc 0.832, macro-F1 0.803; threat QWK 0.771; route F1 0.675; condition AUROC 0.821
End-to-end (Stage 1 → Stage 2)	test split	material acc 0.840 vs 0.840 oracle; route F1 0.724 vs 0.740; detection rate 1.0

4.6 External Validation

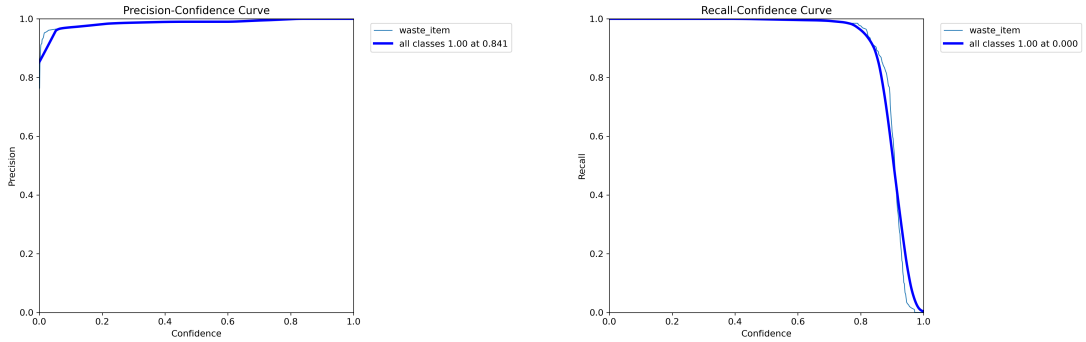
To assess generalization beyond the in-house data, material classification is evaluated on public TrashNet [8] and the detector is stress-tested on TACO [17]. The TrashNet experiments use a ResNet50 material classifier. Zero-shot transfer reaches accuracy 0.695 and macro-F1 0.632 on 476 images when predictions are restricted to the four overlapping classes (glass, metal, paper, plastic); using the full ten-way output gives accuracy 0.439. A separate six-class fine-tuning experiment reaches accuracy 0.889 and macro-F1 0.861 on 503 test images; this result is exploratory because checkpoint selection used the evaluation split. Figure 4.10 shows both results, with their class spaces and test subsets identified, so the difference is interpreted as two transfer settings rather than a paired gain.

TACO's cluttered multi-object scenes lie outside the single-item capture setting used for detector training. On 300 TACO images, mAP@50 is 0.096, mAP@50-95 is 0.045, precision is 0.299, and recall is 0.145. This substantial reduction identifies



(a) Precision–Recall

(b) F1–Confidence



(c) Precision–Confidence

(d) Recall–Confidence

Figure 4.8: Earlier YOLO11 detector evaluation: precision–recall, F1–confidence, precision–confidence, and recall–confidence curves.

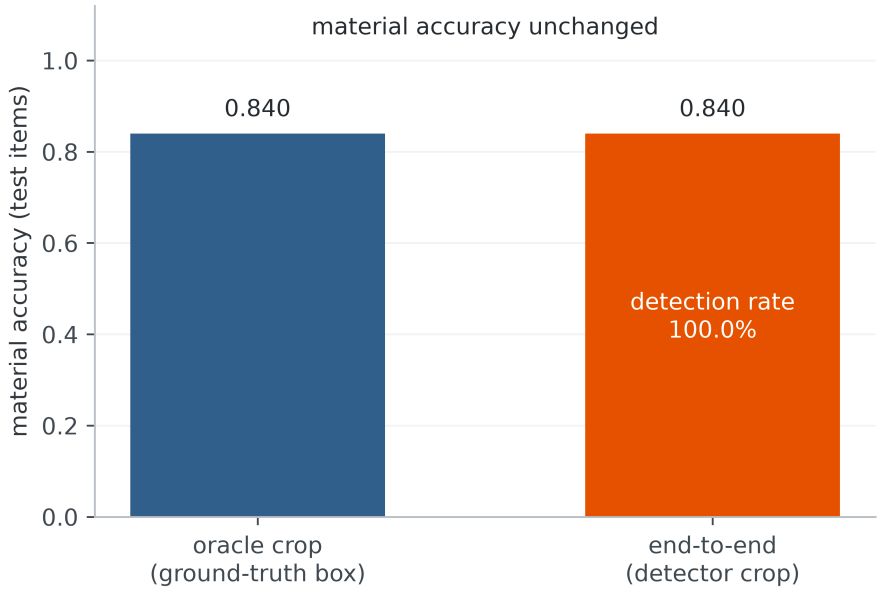


Figure 4.9: End-to-end material accuracy versus an oracle crop.

a generalization boundary of the current system and motivates broader scene data and further detector adaptation. The external experiments test material recognition and localization; evaluation of the custom threat and routing labels remains an internal study.

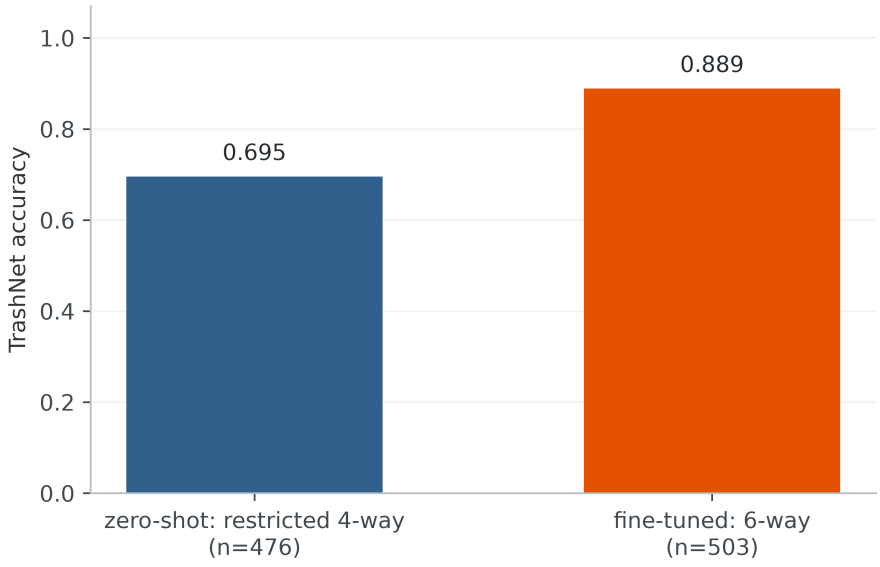


Figure 4.10: Material transfer to TrashNet under restricted four-way zero-shot and six-way fine-tuned evaluation.

4.7 Multi-View Aggregation

An item is photographed from several angles, and its final prediction must combine the evidence from those views. The default averages logits before applying the output activation. Table 4.6 compares this choice with six alternative parameter-free rules on the same 457 cross-validated items, and Figure 4.11 visualizes their accuracy and threat-sensitivity trade-offs. For this comparison, threat is decoded as the most probable CORN class, whereas the headline evaluation uses the ordinal median obtained by counting cumulative probabilities above 0.5. This explains the mean-logit QWK of 0.778 here versus 0.771 in Table 4.2. Threat-threshold recall is measured over the 327 items with true threat ≥ 2, and false-positive rate over the remaining 130 items; these measures concern the threat head rather than the final disposal decision.

Mean logits gives the highest material macro-F1 (0.803), while median logits gives the highest threat QWK (0.783). Best-view selection raises route F1 to 0.697 but reduces material F1 to 0.779. Max/noisy-OR is the most sensitive to threat, raising threshold recall from 0.933 to 0.988, but its false-positive rate increases from 0.108 to 0.269 and QWK falls to 0.732. Log-sum-exp offers an intermediate

Table 4.6: Multi-view aggregation on 457 cross-validated items, with mode threat decoding. Threat recall and FPR use threshold ≥ 2; best per column in bold.

Aggregation	mat F1	route F1	QWK	thr. rec.	thr. FPR
mean-logit (default)	**0.803**	0.675	0.778	0.933	**0.108**
mean-prob	0.795	0.691	0.778	0.930	**0.108**
conf.-weighted	0.798	0.692	0.773	0.927	**0.108**
best-view	0.779	**0.697**	0.771	0.927	**0.108**
median-logit	0.801	0.669	**0.783**	0.942	**0.108**
log-sum-exp	0.785	0.694	0.760	0.982	0.208
max / noisy-OR	0.778	0.694	0.732	**0.988**	0.269

recall/false-alarm trade-off. The default mean-logit rule therefore provides a useful balance for joint prediction, while a more sensitive rule may be appropriate when missing a high-threat item carries a greater cost than unnecessary special handling.

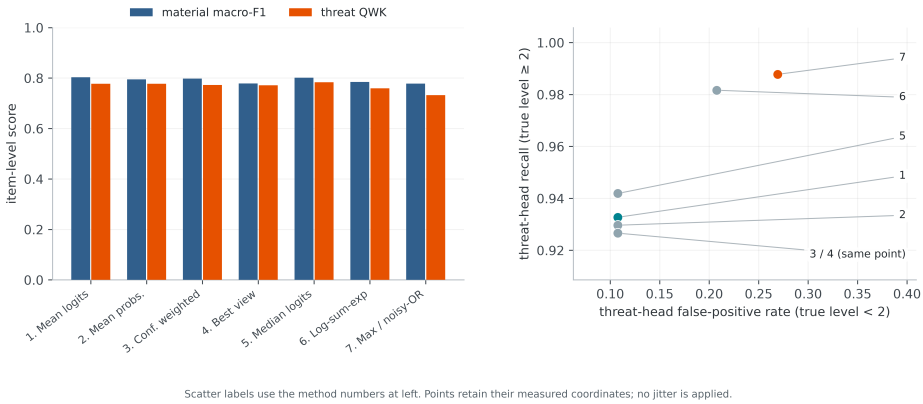


Figure 4.11: Multi-view aggregation: material macro-F1 and threat QWK (left), and threat recall versus false-positive rate at level ≥ 2 (right). Numbered labels identify the aggregation methods.

4.8 Interpretability and Qualitative Results

Aggregate metrics are complemented by Grad-CAM saliency maps and end-to-end walkthroughs, which make the model's behavior on individual items easier to inspect.

Figure 4.12 shows six representative examples, with the original image, material-head saliency, and threat evidence arranged by column. Battery, glass, and sharps are recognized with displayed view confidences of 0.98, 1.00, and 0.99.

The correctly classified composite coffee cup (0.99) draws attention to the cup and printed logo, while some maps also emphasize surrounding regions. The chemical container illustrates a harder case: its plastic appearance leads to a plastic prediction at 0.50 on the displayed view. Across its views, the item-level prediction is plastic with confidence 0.991 and a Recycle decision despite the true Special route. This error shows why visual material alone is insufficient for chemical waste and motivates additional cues such as warning text and hazard pictograms.

Figure 4.13 traces three test items through detection, cropping, classification, multi-view aggregation, and routing. The battery has five views and is assigned threat 3 rather than the true level 2, yet the final Special route is correct. The poor-condition glass item has four views and is correctly sent to Special, while the composite carton has five views and is correctly sent to Landfill. Together, the examples illustrate how condition, material, and threat contribute to disposal decisions, and how a one-level threat error need not change the final route.

4.9 Rare-Class Analysis

Composite and hazardous_chemical are the weakest material classes in the cross-validated evaluation, with F1 values of 0.513 and 0.634 and 21 items each. Table 4.7 shows their main confusions. Composite is most often mistaken for plastic or paper, its visible constituents, while hazardous chemical is confused with the ordinary material of its container. The expanded dataset now contains 34 composite and 36 hazardous-chemical items, but the reported cross-validated results concern the earlier pool. These classes remain priorities for targeted collection and improved recognition of cues that distinguish the item's contents or construction from its outward material.

The decision layer can contain some material errors by using the other task outputs. Among the 21 hazardous-chemical items, 13 are correctly classified by material (recall 0.619), while 15 are routed to Special or referred for review (0.714). This is still below the pooled review-inclusive hazard recall of 0.888 over all 169 Special-route items. The chemical examples routed to Recycle or Landfill show

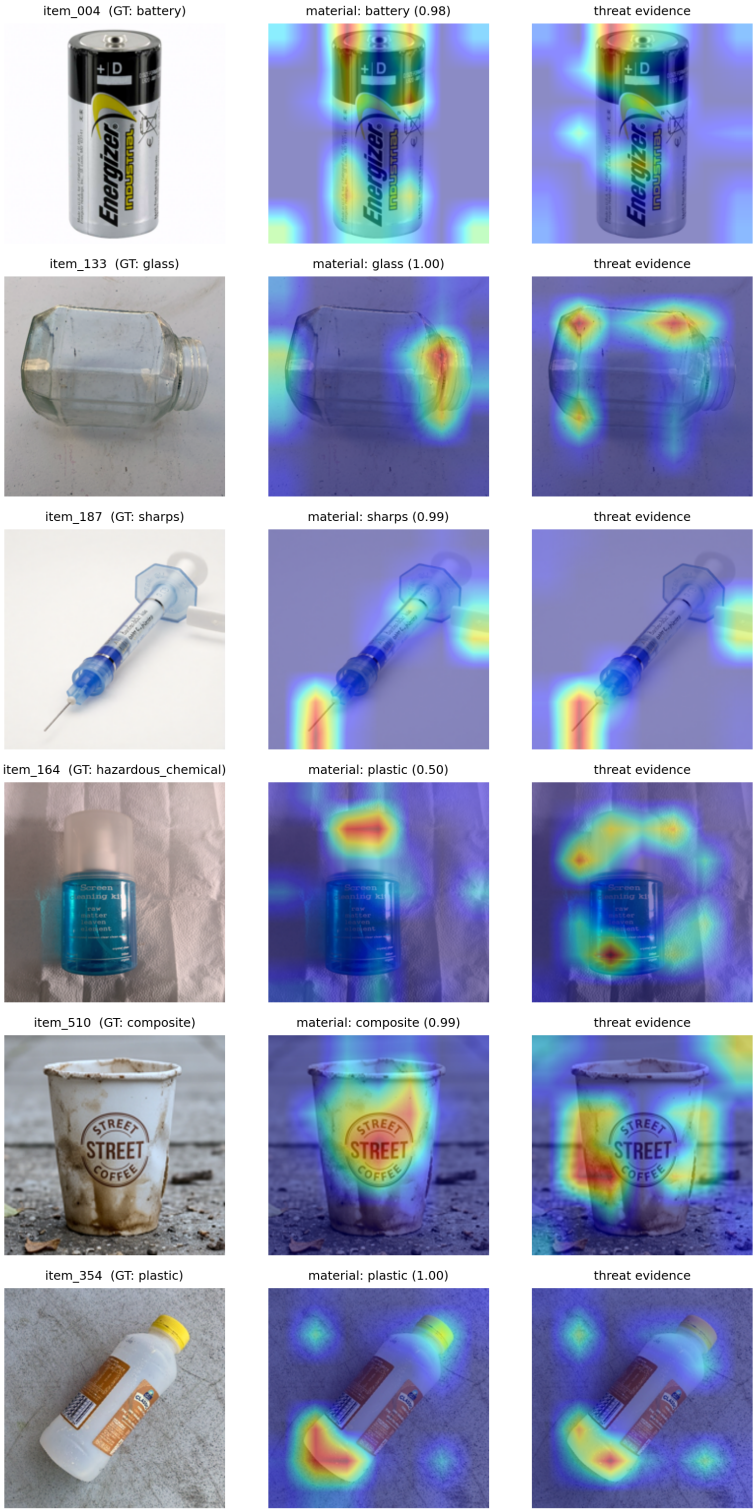


Figure 4.12: Grad-CAM examples: original image, material-head saliency with view confidence, and threat evidence.

the remaining difficulty: a hazardous container can resemble ordinary packaging while attracting a confident prediction. OCR of warning text, hazard pictograms, and resin codes offers a promising direction for addressing these errors.

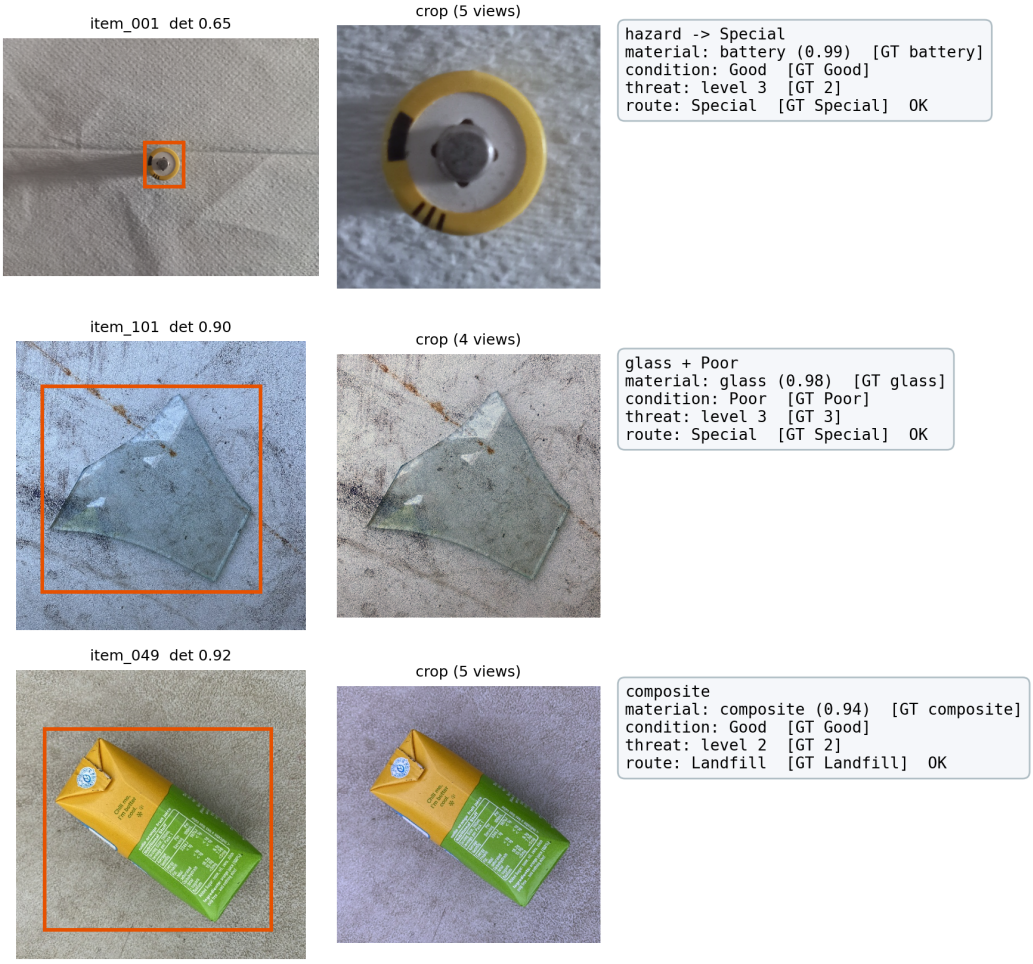


Figure 4.13: End-to-end pipeline walkthrough: detection, crop, and the final routing decision.

Table 4.7: Cross-validated confusion for the two weakest material classes.

True class	Correct	Confused as (counts)
hazardous_chemical (n=21)	13	plastic 3, metal 2, glass 1, battery 1, composite 1
composite (n=21)	10	plastic 6, paper 3, e_waste 2

4.10 Discussion and Comparison with Related Work

Table 4.8 positions the proposed system against the reference points reviewed in Chapter 2. The comparison provides context across material classification, detection, routing, and hazardous handling; differences in datasets, label spaces, and metrics must be considered when interpreting the values.

Table 4.8: Comparison with published reference points.

Task	**Published reference**	**This work**
Material classification	Garbage-12: EfficientNet-B0 86%, ResNet50 ∼88% [9]	acc 0.832 / macro-F1 0.803 (457-item pooled CV)
Material (clean studio)	TrashNet: GoogLeNet 97.9%, ResNet50+SVM 87% [8]	fine-tuned acc 0.889 (6 classes, 503 images); zero-shot restricted acc 0.695 (4 classes, 476 images)
Waste detection (mAP)	YOLOv8-based ∼89.5% mAP [15]	YOLO11 mAP@50 0.992 / mAP@50-95 0.853
Recyclability / route	MHS >90% (binary recyclable/other) [20]	4-class route acc 0.759 / macro-F1 0.675 (different target)
Hazardous handling	flat 4-class incl. hazardous ∼65.8% [21]	threat QWK 0.771; strict Special recall 0.876, or 0.888 including review

Three aspects distinguish the experimental setting. First, task granularity is greater: the system predicts ten materials and four disposal routes alongside condition and ordinal threat, whereas several reference systems address material or binary recyclability alone. Second, evaluation is performed at the physical-item level, keeping different views of an item in one split, and includes both held-out and cross-validated results. Third, a single shared backbone supports all four outputs; the material-only control reaches the same material score on the held-out split. External TrashNet evaluation complements the internal experiments, while the low TACO detection result shows the need for further adaptation to cluttered scenes. The contribution is therefore the joint treatment of material, condition, route, and ordered threat, together with a decision layer that connects these predictions to handling actions.

4.11 Conclusion

The selected ConvNeXt-S reaches material accuracy and macro-F1 of 0.840 on 50 held-out items, while pooled cross-validation over 457 items gives accuracy 0.832 and macro-F1 0.803. Detector crops preserve the held-out material result, with

small changes in route and condition performance. Strict Special-routing recall is 0.876 in the cross-validated decision evaluation, rising to 0.888 when review referrals count as safe diversion. The experiments also identify the principal areas for improvement: rare-material recognition, route imbalance, per-head calibration, and generalization to cluttered scenes. The next chapter concludes the thesis.

Chapter 5

Conclusion

5.1 Conclusion

This thesis developed a threat-aware, two-stage waste detection and classification system that operates on real-world-condition images. A single-class YOLO11s detector localizes the waste item, and a shared-backbone network with four task heads jointly predicts material (ten classes), condition, disposal route, and an ordinal threat level; the heads are fused by a regulation-grounded decision layer with a threat-based safety backstop, and per-head calibration is evaluated. Among ten classifier configurations spanning convolutional and transformer backbones and an ensemble, ConvNeXt-S (ImageNet-22k) was the strongest individual backbone on the primary material task, and the system was evaluated by cross-validation over 457 items with bootstrap confidence intervals.

The cross-validated results are strong and honest: material accuracy 0.832 (95% confidence interval 0.797–0.864) and macro-F1 0.803, with material calibration error 0.036, threat QWK 0.771, and condition AUROC 0.821; the held-out test model reaches material accuracy 0.840 and macro-F1 0.840 on 50 items. The detector reaches mAP@50 0.992, and end-to-end detection-plus-classification shows no measurable material-accuracy loss versus an oracle crop on those 50 items, at a detection rate of 1.0, although condition AUROC and route macro-F1 decrease slightly. Above all, the routing policy sends 148 of 169 hazardous items to Special (recall 0.876). Including two referrals for human review raises safety recall to 0.888 (95% confidence interval 0.838–0.933) at coverage 0.961, showing both the value of safety-aware routing and the remaining missed hazards. The principal contribution is modelling disposal threat as a rank-consistent ordinal output

coupled to a regulation-grounded decision, with calibration, external validation, and rare-class analysis providing a broader assessment of the system.

5.2 Limitations and Threats to Validity

The work has clear, openly stated limitations. The dataset comprises 485 items, and composite and hazardous_chemical remain the hardest material classes. The cross-validated analysis includes 21 items from each class, with F1 scores of 0.513 and 0.634 respectively; limited support and visual ambiguity make reliable recognition difficult. The threat backstop can intercept some material errors, but it does not contain every hazardous miss. The route head is the weakest head (cross-validated macro-F1 0.675), dragged down by the very small Reuse class and by Recycle/Landfill confusion. Finally, the system is appearance-only, limiting its ability to distinguish visually similar classes. Threat level 0 has no evaluation examples, so performance at that level remains unvalidated.

Three threats to validity are acknowledged explicitly. Internal validity: the held-out test split contains 50 items, so its point estimates carry wide implicit intervals; this is mitigated by reporting cross-validation over 457 items with bootstrap confidence intervals and by paired significance tests for the backbone choice. External validity: all items were photographed in one geographic and equipment context, and the TACO transfer result (mAP@50 0.096) quantifies how far the detector generalizes beyond single-item photographs; conclusions therefore apply to single-item capture settings, not to cluttered street scenes. Construct validity: the ordinal threat labels follow the published Litter Impact Index rubric rather than ad-hoc judgement, but any rubric involves interpretation at class boundaries; the routing matrix pins the label definitions to cited regulations to keep this subjectivity bounded and auditable.

5.3 Future Work

Several directions follow naturally from these limitations:

- **Multi-modal disambiguation.** Adding a second signal such as optical character recognition of hazard pictograms, warning text (“danger,” “poison”), and resin/recycling codes is a promising way to separate visually identical classes such as chemical versus ordinary containers, and to improve the rare classes.
- **Larger and broader data.** Collecting more items, especially of the scarce hazardous and composite streams and across more environments, could improve performance and the reliability of rare-class evaluation.
- **Stronger route modelling.** The route head could be improved with route-specific features or by explicitly conditioning on the (calibrated) material and condition predictions.
- **Deployment study.** A field trial on an edge device or sorting station, measuring throughput, latency, and the value of the abstain option with a human in the loop, would assess real-world readiness.
- **Temporal and video input.** Using multiple frames or short video of an item could resolve ambiguities that a single still cannot.

Together, these would extend a complete, safety-aware pipeline toward broader, real-world deployment.

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